
DomED: Redesigning Ensemble Distillation for Domain Generalization

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Abstract

Domain generalization aims to improve model performance on unseen, out-of-distribution (OOD) domains, yet existing methods often overlook the crucial aspect of uncertainty quantification in their predictions. While ensemble learning combined with knowledge distillation offers a promising avenue, naively combining these techniques is non-trivial and remains largely unexplored in the context of domain generalization. In this work, we systematically investigate different ensemble and distillation strategies for domain generalization tasks and design a tailored data allocation scheme. This approach trains teacher models on distinct subsets of domains and performs distillation on complementary (unseen) subsets, thereby fostering model diversity and training efficiency. Moreover, our theoretical analysis demonstrates that distilling from teachers on unseen domains effectively filters out domain-specific spurious correlations. To address the accuracy degradation often observed with standard uncertainty distillation, we further develop a novel technique that decouples uncertainty distillation from the standard distillation process, enabling accurate uncertainty estimation without compromising model accuracy. Our proposed method, *Domain-aware Ensemble Distillation* (DomED), is extensively evaluated against state-of-the-art domain generalization and ensemble distillation techniques across multiple benchmarks, achieving competitive accuracies and substantially improved uncertainty estimates.

1. Introduction

A fundamental assumption in many machine learning techniques is that training and test data are drawn from the same

distribution. However, this assumption often fails in real-world scenarios, where models trained in one environment may be deployed in a different environment, leading to a distributional shift. Domain generalization (DG) (Shankar et al., 2018; Zhou et al., 2020) addresses this challenge by training a model on multiple source domains such that it can better generalize to unseen target domains.

Existing DG methods primarily focus on learning domain-invariant representations, employing techniques like explicit feature alignment (Ghifary et al., 2015; Maniayar et al., 2020), domain adversarial learning (Du et al., 2021; Li et al., 2018b), and feature disentanglement (Mahajan et al., 2021; Zhang et al., 2022). While effective, these methods still exhibit limited generalization performance on out-of-distribution (OOD) data, often not significantly outperforming carefully-tuned empirical risk minimization (ERM) (Gulrajani & Lopez-Paz, 2020). More critically, they tend to produce overconfident yet erroneous predictions on OOD data (Ovadia et al., 2019), rendering their predictions unreliable. This highlights the need to consider both prediction accuracy and uncertainty estimation in developing robust DG methods.

Ensemble methods, which average the outputs of multiple models, are known to improve generalization and provide more accurate uncertainty estimates (Opitz & Maclin, 1999; Dietterich, 2000; Zhou et al., 2002; Rokach, 2010; Lakshminarayanan et al., 2017), particularly for epistemic uncertainty. However, the computational and memory overhead of using ensembles at inference time can be prohibitive. Knowledge distillation offers a solution by compressing an ensemble into a single, efficient model while preserving its uncertainty estimation capabilities (Tran et al., 2020; Malinin et al., 2019; Ferienc & Rodrigues, 2022). However, two critical hurdles remain in adapting them to DG. First, standard ensembles require training multiple models on all data, which is costly, while self-distillation from a single model lacks the diversity required for robust uncertainty estimation (Lee et al., 2022). Second, we empirically find that naively applying standard uncertainty distillation (e.g., minimizing Dirichlet NLL) to DG tasks degrades classification accuracy. Consequently, a principled framework should simultaneously optimize for computational efficiency, model diversity, and the balance between prediction accuracy and uncertainty, which existing methods fail to achieve.

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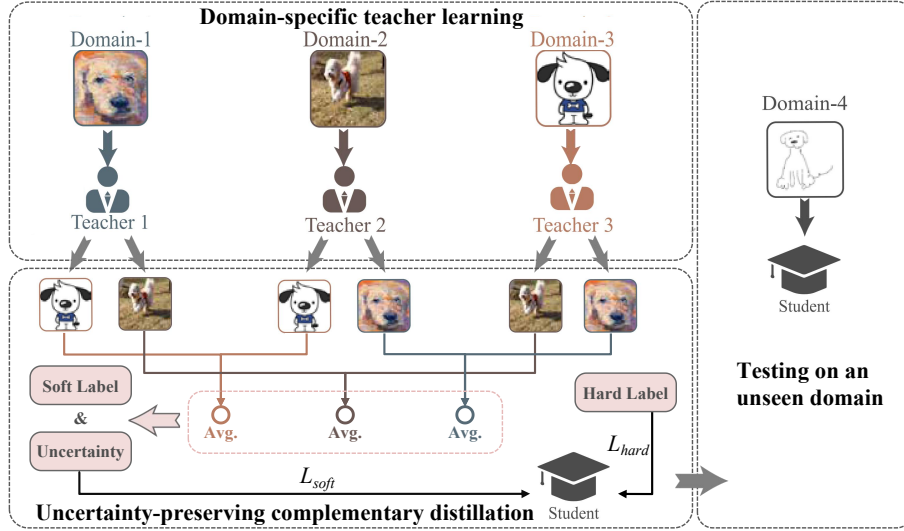


Figure 1. Illustration of DomED. Each teacher model is trained on a specific source domain. The teachers collectively make predictions on data complementary to their respective training data, such that their generalization ability along with their inherent uncertainty information can be distilled to a student model. After distillation, the student model is evaluated on a previously unseen target domain.

In this work, we develop a tailored ensemble learning and knowledge distillation scheme for domain generalization. Recent work suggests that training multiple models to exploit the “multi-view” structure in data is crucial for the success of ensemble methods (Allen-Zhu & Li, 2020). This aligns with the understanding that model diversity is key in ensemble learning (Brown et al., 2005; Nam et al., 2021a). Unlike regular classification tasks, DG provides domain labels for each data sample, presenting a natural opportunity for multi-view learning. We therefore propose to train the base models of an ensemble on different, non-overlapping subsets of domains, and subsequently perform distillation on unseen domains. This strategy enhances model diversity and suppresses spurious features while reducing the training cost of individual models. We also evaluate and compare different data allocation schemes for ensembling and distillation to identify the optimal scheme. To the best of our knowledge, we are the first to systematically investigate the possible data allocation schemes for adapting ensemble distillation to the particular setting of domain generalization.

Beyond achieving high accuracy on unseen domains, our goal is to distill the uncertainty estimation capability of ensembles. A common approach is to train a prior network to output a conjugate prior (e.g., a Dirichlet distribution for classification tasks) that captures the output distributions of the base models (Malinin & Gales, 2018; Malinin et al., 2019). However, this approach often leads to significantly lower model accuracy than standard distillation (Hinton et al., 2015). To address this, we introduce a novel technique that decouples uncertainty distillation from standard distillation, allowing for accurate model predictions and uncertainty estimates simultaneously. We refer to our ap-

proach as *Domain-aware Ensemble Distillation* (DomED), as illustrated in Figure 1.

Our main contributions are summarized as follows:

- We explore tailored ensemble and distillation strategies for domain generalization tasks, and develop a novel data allocation scheme that trains and distills teacher models on complementary domains, which enhances model diversity and training efficiency. We theoretically show that the complementary distillation strategy allows the student to discard domain-specific spurious features that individual teachers memorize.
- We identify that the commonly used uncertainty distillation method degrades the accuracy of an ensemble after distillation in the context of DG. We address this by proposing a decoupled distillation technique that preserves both mean prediction accuracy and model uncertainty.
- We conduct extensive experiments to compare different data allocation schemes and evaluate our approach, DomED, on multiple domain generalization benchmarks. Our results demonstrate that DomED achieves competitive accuracies and significantly improved uncertainty quantification compared to existing methods.

2. Methods

We present our ensemble learning and knowledge distillation methods for domain generalization. First, we formally define the domain generalization problem and introduce the relevant notation. We then discuss various ensembling

and distillation strategies and develop a tailored scheme for domain generalization that achieves high model accuracy and reduced training cost. Based on this scheme, we further propose a novel uncertainty-preserving distillation method, which decouples uncertainty distillation from the standard distillation process, to simultaneously achieve accurate model predictions and uncertainty estimates.

2.1. Problem Definition

We consider a domain generalization problem with M source domains (training domains), whose union is denoted as $\mathcal{D}_{tr} = \bigcup_{m=1}^M \mathcal{D}_{tr}^m$, and one unseen target domain (test domain), $\mathcal{D}_{te}$. Each source domain, $\mathcal{D}_{tr}^m$, contains N_m independent and identically distributed (i.i.d.) labeled training samples, i.e., $\mathcal{D}_{tr}^m := \{(x_i^m, y_i^m)\}_{i=1}^{N_m}$. Similarly, the target domain comprises N_{te} unlabeled i.i.d. samples, $\mathcal{D}_{te} := \{x_j\}_{j=1}^{N_{te}}$. While all domains share a common feature space $\mathcal{X}$ and label space $\mathcal{Y}$, the core challenge is that the data distribution varies across domains. In image classification, for instance, domains might represent different visual styles (e.g., photos, sketches) but share the same set of object categories. The objective is to learn a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ using only data from the source domains that can effectively generalize to the unseen target domain.

2.2. Ensemble and Distillation Strategies

In a regular supervised learning problem that has i.i.d. data for both training and testing, the base models of an ensemble are usually trained on the same dataset, and model diversity is introduced only by the independent initialization of their parameters (Allen-Zhu & Li, 2020). Moreover, the distillation of the ensemble is also done on the same set of training data. In contrast, a domain generalization task splits data into multiple domains, making it possible to increase model diversity by training base models on different subsets of domains. Furthermore, for each base model, it is also possible to use a different set of data than its training data for distillation. As shown in Figure 1 and Figure 2, we consider six representative training data allocation schemes, and compare them empirically in Section 3. Among them, Figure 2(a) depicts the self-distillation scheme, and Figure 2(b) depicts the regular ensembling and distillation scheme. For fair comparison, we assume an ensemble of M base models (one for each source domain), $\{T^m\}_{m=1}^M$. In this section, we discuss in detail the scheme employed by DomED (as shown in Figure 1) due to its superior empirical performance and relatively low computational cost. In the following, we also refer to the base models of an ensemble and the distilled model, respectively, as teacher models and student model.

Domain-specific teacher models. The data allocation scheme of DomED is designed to maximize teacher diversity by training each of the M teacher models on a single,

distinct source domain. This strategy treats each domain as a unique “view” of the data, encouraging the models to develop different specializations. Specifically, each teacher model, T^m , is a neural network parameterized by weights θ_T^m and is trained exclusively on its corresponding domain, $\mathcal{D}_{tr}^m$. The complete set of teacher parameters is denoted as $\theta_T = \{\theta_T^m\}_{m=1}^M$. For classification tasks, each teacher is trained to minimize the cross-entropy loss on its domain:

$$\mathcal{L}_T^m = \mathbb{E}_{(x_i^m, y_i^m) \in \mathcal{D}_{tr}^m} [\text{CE}(\pi(x_i^m; \theta_T^m), y_i^m)], \quad (1)$$

where $\text{CE}(\cdot, \cdot)$ denotes the cross-entropy between two categorical distributions, y_i^m denotes the ground-truth label of x_i^m , and $\pi(x_i^m; \theta_T^m)$ represents the predictive distribution output by the teacher.

A key advantage of this domain-specific training scheme is its computational efficiency. While training M teachers may seem costly, the total computational cost can be significantly less than training a single model on all source domains. This is because each domain-specific teacher can be trained with a smaller batch size (e.g. $1/M$) and requires significantly fewer training steps to become an effective guide for distillation. Crucially, as detailed in Appendix B.1, we empirically find that the student model’s performance saturates even before the individual teachers have fully converged, substantially reducing the training overhead that typically makes ensemble methods impractical.

Complementary Distillation. As illustrated in Figure 1, DomED employs a complementary distillation strategy designed to transfer the generalization ability of the teacher models to the student. The core principle is to generate distillation targets for samples from a given source domain, $\mathcal{D}_{tr}^m$, using only the teachers that were never trained on it. Specifically, for an input sample $x_i^m \in \mathcal{D}_{tr}^m$, we first gather the output logits $z^n(x_i^m)$ from all complementary teachers T^n (where $n \neq m$). These logits are then aggregated by averaging to produce a single soft target for distillation:

$$\bar{z}(x_i^m) = \frac{1}{M-1} \sum_{\substack{n=1 \\ n \neq m}}^M z^n(x_i^m). \quad (2)$$

The student model is then trained to emulate these aggregated predictions by minimizing a temperature-scaled cross-entropy loss. For each source domain $\mathcal{D}_{tr}^m$, this soft-target loss is defined as:

$$\mathcal{L}_{\text{soft}}^m = \mathbb{E}_{(x_i^m, \cdot) \in \mathcal{D}_{tr}^m} [\text{CE}(\pi(x_i^m; \theta_S), \bar{\pi}(x_i^m; \theta_T, \tau^m))], \quad (3)$$

where $\pi(x_i^m; \theta_S)$ is the distribution predicted by the student model, and $\bar{\pi}(x_i^m; \theta_T, \tau^m)$ is the distribution derived from the aggregated logits $\bar{z}(x_i^m)$ by applying the softmax function with a temperature hyperparameter τ^m . τ^m is

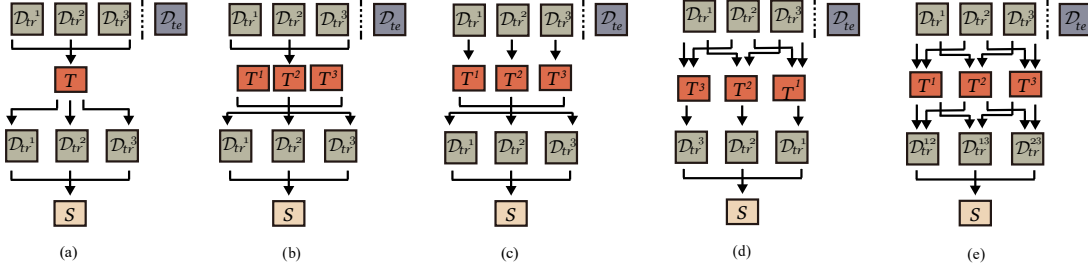


Figure 2. Comparison of different data allocation schemes for ensemble learning and knowledge distillation. Note that these are different from the DomED scheme presented in Figure 1. Given M training domains, (a) self-distillation from a single teacher model; (b) standard ensemble distillation (all teachers trained and distilled on all domains); (c) includes M single-domain teachers, distilled on all domains; (d) includes M teachers, each trained on $M - 1$ domains and distilled on the remaining domain; and (e) includes M teachers, each trained and distilled on the same $M - 1$ domains.

usually set between 1 and 4, and a larger τ produces a softer probability distribution. Note that $\hat{\pi}(x_i^m; \theta_T, \tau^m)$ does not receive gradient and θ_T does not update during distillation. The complementary distillation strategy is also employed in an alternative data allocation scheme shown in Figure 2(d).

Theoretical Motivation. Why does distilling from complementary domains generalize better than self-distillation? We posit that this strategy acts as a filter for spurious correlations. In a domain generalization setting, teachers often learn domain-specific shortcuts (spurious features) to minimize training error. However, a teacher trained on domain A will perceive the spurious features of domain B effectively as noise, as it has not learned to correlate them with the label. Consequently, when we aggregate predictions from teachers who have not seen the current domain (Equation (2)), they rely primarily on invariant features, while their responses to the domain-specific spurious features average out. By mimicking this ‘cleaned’ target, the student learns to ignore the spurious shortcuts present in the input data. We provide a formal analysis for this spurious feature filtering mechanism using a linear structural equation model in Appendix A.

As a common practice for knowledge distillation (Hinton et al., 2015), the student model is also trained on the ground-truth labels to achieve higher model accuracy. Similar to Equation (1), we minimize the cross-entropy loss:

$$\mathcal{L}_{\text{hard}}^m = \mathbb{E}_{(x_i^m, y_i^m) \in \mathcal{D}_{\text{tr}}^m} [\text{CE}(\pi(x_i^m; \theta_S), y_i^m)]. \quad (4)$$

Finally, the two losses in Equation (3) and Equation (4) are combined as the training loss of student model:

$$\mathcal{L}_S = \frac{1}{M} \sum_{m=1}^M [\lambda \mathcal{L}_{\text{soft}}^m + (1 - \lambda) \mathcal{L}_{\text{hard}}^m], \quad (5)$$

where λ is a hyperparameter that balances the soft and hard targets. See Appendix G.1 for more details on tuning λ .

2.3. Uncertainty-preserving Distillation

While the distillation scheme discussed in Section 2.2 produces an accurate student model, it discards the ensemble’s valuable epistemic uncertainty. This limitation is inherent to the standard cross-entropy objective, which trains the student to match only the *mean* of the teacher predictions. In doing so, it collapses the full predictive distribution into a single point estimate, losing the diversity across teacher outputs that signals model uncertainty. To preserve this crucial information, we distill the ensemble’s uncertainty by training the student model to output the parameters of a Dirichlet prior.

Learning a Dirichlet Prior. The Dirichlet distribution is the conjugate prior of the categorical distribution, making it well-suited for modeling the predictive uncertainty of an ensemble. It provides a principled way to represent a distribution over distributions, which naturally captures the set of predictions from the teacher models.

Given C classes, the Dirichlet distribution is defined by its positive concentration parameters $\alpha = [\alpha_1, \dots, \alpha_C]$. The sum of these parameters, $\alpha_0 = \sum_{c=1}^C \alpha_c$, is known as the precision of the distribution; a higher precision indicates lower uncertainty (a more peaked distribution). The probability density function (PDF) is given by:

$$\text{Dir}(\pi; \alpha) = \frac{\Gamma(\alpha_0)}{\prod_{c=1}^C \Gamma(\alpha_c)} \prod_{c=1}^C (\pi_c)^{\alpha_c - 1}, \quad (6)$$

where π is a probability vector. To enable a student model to output this distribution, we parameterize the concentration parameters from its logits, $z_c(x_i^m)$, for an input x_i^m as $\alpha_c(x_i^m) = e^{z_c(x_i^m)}$. The student is then trained to find the Dirichlet parameters that best explain the collection of teacher predictions by minimizing the following negative

log-likelihood (NLL) loss (Malinin et al., 2019):

$$\mathcal{L}_{\text{Dir}} = \mathbb{E}_{(x_i^m, \cdot) \in \mathcal{D}_{tr}^m} \left[\sum_{c=1}^C \ln \Gamma(\alpha_c(x_i^m)) - \ln \Gamma(\alpha_0(x_i^m)) \right. \\ \left. - \frac{1}{M-1} \sum_{\substack{n=1 \\ n \neq m}}^M \sum_{c=1}^C (\alpha_c(x_i^m) - 1) \ln \pi_c(x_i^m; \theta_T^n) \right], \quad (7)$$

where $\pi_c(x_i^m; \theta_T^n)$ is the probability assigned to class c by teacher T^n .

Decoupled Uncertainty Distillation. While directly minimizing the Dirichlet NLL loss ($\mathcal{L}_{\text{Dir}}$) is a common approach for uncertainty distillation (Malinin & Gales, 2018; Malinin et al., 2019), we find that it can degrade accuracy in the challenging context of domain generalization. This issue arises because $\mathcal{L}_{\text{Dir}}$ implicitly attempts to solve two coupled problems simultaneously: matching the *mean* of the ensemble’s predictions and matching their *spread* (i.e., uncertainty). As noted by Ryabinin et al. (2021), these objectives can conflict, forcing the model to compromise on accuracy to better fit the uncertainty.

To resolve this, we propose a novel decoupling strategy that treats mean prediction and uncertainty quantification as orthogonal learning objectives. Unlike prior works that attempt to balance these within a single likelihood term, we assign a specialized loss to each task:

1. We use the standard distillation loss, $\mathcal{L}_S$, to distill the mean prediction. This cross-entropy-based loss is highly effective at aligning the student with the teachers’ mean prediction, preserving model accuracy.
2. We use the Dirichlet NLL loss, $\mathcal{L}_{\text{Dir}}$, to capture the ensemble’s prediction uncertainty. When teachers agree, the target Dirichlet distribution is sharp, teaching the student to be confident; when they disagree, it is flat, teaching the student to reflect this uncertainty.

This decoupling allows the standard cross-entropy term ($\mathcal{L}_S$) to enforce an accurate mean prediction, while the Dirichlet term ($\mathcal{L}_{\text{Dir}}$) focuses on shaping the distribution’s spread. This leads to our final, decoupled objective:

$$\mathcal{L}'_S = \mathcal{L}_S + \beta \mathcal{L}_{\text{Dir}}, \quad (8)$$

where β is a hyperparameter balancing the two terms. In practice, we set β to a small value (e.g., 0.01), effectively using $\mathcal{L}_{\text{Dir}}$ as a regularizer that fine-tunes the model’s uncertainty without disrupting the primary accuracy signal from $\mathcal{L}_S$. As we show in our analyses of the loss components (Appendices F and J) and the impact of β on convergence (Appendix G.2), this conceptually simple decoupling strategy successfully preserves accuracy while enabling robust uncertainty transfer.

3. Experiments

Our experiments systematically evaluate DomED on several domain generalization benchmarks. We first systematically compare the performance of six data allocation schemes proposed in this work. We then benchmark DomED against state-of-the-art methods on DomainBed and assess its uncertainty quantification performance using standard metrics. In addition, we present an ablation study of the distillation loss in Appendix F, and verify the architectural robustness of DomED in Appendix K.

3.1. Experimental Setup

Datasets and evaluation protocol. We use the following datasets in our experiments: 1) PACS (Li et al., 2017), a widely used multi-source domain generalization dataset comprising 9,991 images across 7 classes and 4 domains (Art_painting, Cartoon, Photo, and Sketch), 2) OfficeHome (Venkateswara et al., 2017), a dataset of 15,500 images from 65 classes and 4 domains (Art, Clipart, Product, and Real World), 3) VLCS (Fang et al., 2013), which contains 10,729 images of 5 classes sourced from Caltech101, LabelMe, SUN09, and VOC 2007, 4) TerraIncognita (Beery et al., 2018), a slightly larger dataset of 24,788 images from 10 classes and 4 domains, and 5) DomainNet (Peng et al., 2019), a significantly larger dataset of 586,575 images from 345 classes and 6 domains. Following the evaluation protocol of DomainBed (Gulrajani & Lopez-Paz, 2020), we perform leave-one-domain-out evaluation and use training-domain validation for hyperparameter tuning. In Table 1 and Table 2, the reported accuracies are average of 7 runs.

Implementation Details. To ensure a fair comparison, we follow the standard protocol of DomainBed (Gulrajani & Lopez-Paz, 2020) for all experiments. We use a pre-trained ResNet-50 (He et al., 2016) backbone. For PACS, TerraIncognita, and DomainNet datasets, we use Adam optimizer (Kingma & Ba, 2014) with a learning rate of 5×10^{-5} . For OfficeHome and VLCS, we use stochastic gradient descent (SGD) with an initial learning rate of 2×10^{-2} and a momentum of 0.9. Other hyperparameters, such as batch size, dropout rate, and weight decay, are adopted from recent state-of-the-art methods (Cha et al., 2021; 2022).

3.2. Domain Generalization

Data allocation schemes. In addition to the complementary data allocation scheme illustrated in Figure 1, we investigate five other schemes as shown in Figure 2. Among them, scheme (a) is simple self-distillation without ensembling, scheme (b) corresponds to regular ensembling and distillation that use the full training data for both purposes, and the other three can be considered as variants of DomED. Compared to DomED, scheme (c) distills from all teachers

Table 1. Comparison of different training data allocation schemes for ensemble learning and knowledge distillation. The best and second-best results are indicated by bold and underlined, respectively.

Dataset	Domain	ERM	DomED (Teachers)	Scheme (a)	Scheme (b)	Scheme (c)	Scheme (d)	Scheme (e)	DomED (Ours)
PACS	<i>Art Painting</i>	84.8 $\pm$ 0.2	78.2 $\pm$ 2.8	85.9 $\pm$ 0.5	86.2 $\pm$ 1.3	87.2 $\pm$ 0.5	87.5 $\pm$ 0.8	85.9 $\pm$ 0.6	87.5 $\pm$ 0.5
	<i>Cartoon</i>	80.0 $\pm$ 0.6	69.2 $\pm$ 1.0	81.7 $\pm$ 0.7	80.4 $\pm$ 0.8	81.0 $\pm$ 0.6	82.1 $\pm$ 0.5	80.1 $\pm$ 0.7	<u>81.5</u> $\pm$ 0.3
	<i>Photo</i>	96.2 $\pm$ 0.0	95.0 $\pm$ 0.2	96.9 $\pm$ 0.3	96.7 $\pm$ 0.2	<u>97.0</u> $\pm$ 0.2	96.5 $\pm$ 0.8	96.4 $\pm$ 0.2	97.1 $\pm$ 0.2
	<i>Sketch</i>	79.3 $\pm$ 0.3	68.6 $\pm$ 4.8	79.7 $\pm$ 0.4	79.1 $\pm$ 1.1	<u>81.4</u> $\pm$ 0.4	80.6 $\pm$ 0.2	77.3 $\pm$ 0.8	81.6 $\pm$ 0.4
	Avg.	85.1 $\pm$ 0.2	77.7 $\pm$ 0.9	86.1 $\pm$ 0.4	85.6 $\pm$ 0.4	86.6 $\pm$ 0.3	<u>86.7</u> $\pm$ 0.2	84.9 $\pm$ 0.3	86.9 $\pm$ 0.2
Office Home	<i>Art</i>	61.3 $\pm$ 0.3	64.6 $\pm$ 0.4	67.2 $\pm$ 0.5	67.3 $\pm$ 0.6	67.5 $\pm$ 0.2	<u>67.6</u> $\pm$ 0.2	67.4 $\pm$ 0.3	67.7 $\pm$ 0.1
	<i>Clipart</i>	52.1 $\pm$ 0.4	52.1 $\pm$ 0.2	56.4 $\pm$ 0.6	56.4 $\pm$ 0.6	56.7 $\pm$ 0.3	<u>56.9</u> $\pm$ 0.2	56.7 $\pm$ 0.3	57.1 $\pm$ 0.2
	<i>Product</i>	76.6 $\pm$ 0.3	75.6 $\pm$ 0.3	77.7 $\pm$ 0.2	77.6 $\pm$ 0.3	<u>78.1</u> $\pm$ 0.2	78.0 $\pm$ 0.2	77.7 $\pm$ 0.1	78.2 $\pm$ 0.2
	<i>RealWorld</i>	78.5 $\pm$ 0.2	79.1 $\pm$ 0.2	81.3 $\pm$ 0.1	81.5 $\pm$ 0.2	<u>81.7</u> $\pm$ 0.1	81.6 $\pm$ 0.1	81.2 $\pm$ 0.1	81.8 $\pm$ 0.1
	Avg.	67.1 $\pm$ 0.2	67.8 $\pm$ 0.2	70.6 $\pm$ 0.3	70.6 $\pm$ 0.4	<u>71.0</u> $\pm$ 0.2	<u>71.0</u> $\pm$ 0.1	70.7 $\pm$ 0.2	71.2 $\pm$ 0.2
VLCS	<i>Caltech101</i>	97.8 $\pm$ 0.1	93.8 $\pm$ 3.7	98.2 $\pm$ 0.3	98.3 $\pm$ 0.3	98.5 $\pm$ 0.2	98.2 $\pm$ 0.1	97.8 $\pm$ 0.4	98.5 $\pm$ 0.1
	<i>LabelMe</i>	64.2 $\pm$ 0.3	58.0 $\pm$ 1.5	<u>65.4</u> $\pm$ 0.4	65.5 $\pm$ 0.5	64.8 $\pm$ 0.3	64.9 $\pm$ 0.2	63.6 $\pm$ 0.4	65.3 $\pm$ 0.2
	<i>SUN09</i>	72.6 $\pm$ 0.4	77.0 $\pm$ 0.9	73.3 $\pm$ 0.9	74.8 $\pm$ 0.4	78.0 $\pm$ 0.2	78.8 $\pm$ 0.2	72.9 $\pm$ 0.3	<u>78.4</u> $\pm$ 0.3
	<i>VOC2007</i>	77.6 $\pm$ 0.4	76.7 $\pm$ 1.5	77.3 $\pm$ 0.4	77.7 $\pm$ 0.3	78.9 $\pm$ 0.2	77.8 $\pm$ 0.1	77.2 $\pm$ 0.4	<u>78.4</u> $\pm$ 0.2
	Avg.	78.0 $\pm$ 0.3	76.4 $\pm$ 1.7	78.5 $\pm$ 0.5	79.1 $\pm$ 0.3	80.1 $\pm$ 0.2	79.9 $\pm$ 0.1	77.9 $\pm$ 0.4	80.1 $\pm$ 0.1

regardless of their training domains, scheme (d) allocates each teacher $M - 1$ training domains instead of 1 but also distills on complementary domains, whereas scheme (e) distills on the respective training domains of teachers.

We evaluate the six data allocation schemes along with two baselines, ERM and DomED without distillation, on PACS, OfficeHome, and VLCS. The results are shown in Table 1. Comparing schemes (b) and (c), the latter achieves better accuracies on all three datasets, which suggests that training base models on different subsets of domains can indeed lead to better generalization, possibly due to increased model diversity. By comparing DomED with scheme (c), we also observe that complementary distillation achieves comparable or slightly better accuracies than distillation on all domains, thus corroborating the intuition that DomED can distill the generalization ability into student model. Moreover, despite their similar classification performance, DomED outperforms scheme (c) in terms of uncertainty quantification (see Table 3 for a comprehensive comparison of uncertainty quantification for all schemes). In contrast, distillation only on training domains (scheme (e)) can result in significantly worse accuracies compared to complementary distillation (scheme (d)). Furthermore, while both DomED and scheme (d) employ complementary distillation with comparable accuracies, DomED restricts each teacher to a single domain. This design minimizes training overhead while maximizing model diversity (see Appendix C.1), a prerequisite for robust uncertainty quantification. Finally, we note that the teacher models of DomED are relatively weak as each individual model is trained on only one domain; thus, they perform poorly compared to the distilled model. Nevertheless, the

distilled model proves to be resilient even when individual teachers are trained on scarce data, mitigating the risk of negative transfer (see Appendix D). Collectively, the systematic evaluation confirms that our data allocation strategy is a principled and effective design choice for achieving high teacher diversity and robust distillation.

Results on DomainBed. We evaluate DomED on the DomainBed benchmark against a wide array of state-of-the-art methods, with results presented in Table 2. To provide a clear comparison, we group methods into standard approaches and those that utilize weight averaging techniques like SWAD to further boost performance. As a standalone method, DomED demonstrates strong performance, achieving significant gains over the ERM baseline: +1.8 pp on PACS, +4.1 pp on OfficeHome, +2.1 pp on VLCS, +2.4 pp on TerraIncognita, and +1.7 pp on DomainNet. The improvement is particularly pronounced on OfficeHome, a dataset known for its large inter-domain gap. This setting is ideal for our approach, as it allows domain-specific teachers to become highly specialized, and our complementary distillation can effectively integrate their diverse knowledge to improve generalization. When paired with SWAD, our method achieves a top-tier average accuracy of 67.8%, which is competitive with other leading methods that employ significantly more computation, such as EoA (test-time ensembling) and DiWA (weight averaging from 60 runs). This highlights our efficiency, delivering state-of-the-art performance without significant overhead (see Appendix B for details of computational cost).

Table 2. Classification accuracy (%) on the DomainBed benchmark. Methods are grouped into standalone approaches and those using weight averaging. Best and second-best results in each group are in **bold** and underlined, respectively. “TerraInc.” stands for TerraIncognita. [†]Results reported by Gulrajani & Lopez-Paz (2020). [‡]Results reported by Di Zhao et al. (2025). [§]Test-time ensembling. [¶]Weight averaging from 60 runs.

Method	PACS	OfficeHome	VLCS	TerraInc.	DomainNet	Avg.
ERM (Vapnik, 1999)	85.1±0.2	67.1±0.2	78.0±0.3	47.8±0.6	44.0±0.1	64.4
CORAL [†] (Sun & Saenko, 2016)	86.2±0.3	68.7±0.3	78.8±0.6	47.6±1.0	41.5±0.1	64.5
MMD [†] (Li et al., 2018b)	84.7±0.5	66.3±0.1	77.5±0.9	42.2±1.6	23.4±9.5	58.8
IRM [†] (Arjovsky et al., 2019)	83.5±0.8	64.3±2.2	78.5±0.5	47.6±0.8	33.9±2.8	61.1
Fish (Shi et al., 2021)	85.5±0.3	68.6±0.4	77.8±0.3	45.1±1.3	42.7±0.2	63.9
SagNet [†] (Nam et al., 2021b)	86.3±0.2	68.1±0.1	77.8±0.5	48.6±1.0	40.3±0.1	64.2
SelfReg (Kim et al., 2021)	85.6±0.4	67.9±0.7	77.8±0.9	47.0±0.3	42.8±0.1	64.2
NKD [‡] (Wang et al., 2021)	83.3±0.4	71.1±0.3	77.1±0.3	37.2±0.3	42.4±0.2	62.2
MIRO (Cha et al., 2022)	85.4±0.4	70.5±0.4	79.0±0.0	50.4±1.1	44.3±0.2	65.9
KDDRL (Niu et al., 2023)	86.6±0.8	66.9±1.6	77.3±0.5	48.0±1.1	38.5±0.3	63.4
SAGM (Wang et al., 2023b)	86.6±0.2	70.1±0.2	<u>80.0</u> ±0.3	48.8±0.9	45.0±0.2	66.1
DomainDrop (Guo et al., 2023)	87.9 ±0.3	68.7±0.1	79.8±0.3	51.5 ±0.4	44.4±0.5	<u>66.5</u>
RISE [‡] (Huang et al., 2023)	85.0±0.3	71.5±0.2	77.6±0.2	39.0±0.3	45.2±0.2	63.7
GMDG (Tan et al., 2024)	85.6±0.3	70.7±0.2	79.2±0.3	<u>51.1</u> ±0.9	44.6±0.1	66.3
XDomainMix (Liu et al., 2024)	86.4±0.4	-	-	48.2±1.3	44.4±0.2	-
Arith (Wang et al., 2025)	86.5±0.3	69.4±0.1	79.4±0.3	48.1±1.2	41.5±0.1	65.0
GGA (Ballas & Diou, 2025)	86.4±0.5	67.0±0.3	78.7±0.8	48.5±1.1	44.4±0.2	65.0
BOLD [‡] (Di Zhao et al., 2025)	85.7±0.2	72.6 ±0.2	78.7±0.2	44.3±0.3	46.9 ±0.2	65.6
DomED (Ours)	<u>86.9</u> ±0.2	<u>71.2</u> ±0.2	80.1 ±0.1	50.2±0.5	<u>45.7</u> ±0.1	66.8
SWAD (Cha et al., 2021)	88.1±0.1	70.6±0.2	79.1±0.1	50.0±0.3	46.5±0.1	66.9
CORAL (w/ SWAD) (Cha et al., 2021)	88.3±0.1	71.3±0.1	78.9±0.1	51.0±0.1	46.8±0.0	67.3
SAM (w/ SWAD) (Cha et al., 2021)	87.1±0.2	69.9±0.2	78.5±0.1	45.3±0.9	46.5±0.1	65.5
DNA (w/ SWAD) (Chu et al., 2022)	88.4±0.1	71.2±0.1	79.0±0.1	<u>52.2</u> ±0.4	<u>47.2</u> ±0.1	67.6
EoA [§] (Arpit et al., 2022)	88.6	72.5	79.1	52.3	47.4	68.0
DiWA [¶] (Rame et al., 2022)	89.0	<u>71.6</u>	<u>79.4</u>	49.0	46.3	67.1
DomED (w/ SWAD) (Ours)	88.4±0.2	<u>71.6</u> ±0.2	80.2 ±0.1	51.9±0.3	46.7±0.2	<u>67.8</u>

3.3. Uncertainty Quantification

Beyond strong generalization performance, a crucial advantage of DomED is its ability to provide reliable uncertainty estimates. To evaluate this, we employ four standard metrics: mean classification error (ERR), prediction rejection ratio (PRR) (Malinin et al., 2019), expected calibration error (ECE) (Guo et al., 2017), and negative log-likelihood (NLL). A high PRR is desirable, as it indicates the model can effectively detect and reject its own incorrect predictions. A low ECE signifies well-calibrated confidence. They both are essential for trustworthy OOD generalization.

Calibration and Reliability Analysis. We evaluate DomED against various baselines and data allocation schemes defined in Figure 2 (Schemes (a)–(e)) on the PACS dataset, with full test-time ensembles (a 3-model standard ensemble and EoA (Arpit et al., 2022)) included for reference. The results are summarized in Table 3 (see Appendix L for per-domain results). Standard methods like ERM and CORAL achieve reasonable accuracy but are poorly calibrated, evidenced by high ECE and NLL values.

The poor calibration stems from a rapid entropy collapse during training, whereas the distillation process of DomED retains the dark knowledge from diverse teacher models, resulting in high-entropy predictions (see Appendix E). Post-hoc temperature scaling (Guo et al., 2017) improves calibration but fails to enhance the model’s ability to reject incorrect predictions (PRR). Approximate Bayesian methods like MC Dropout (Gal & Ghahramani, 2016), which require multiple forward passes at inference, prove ineffective in this DG context, degrading accuracy without improving calibration. Furthermore, EnD², while designed for uncertainty distillation, performs poorly due to the absence of ground-truth labels in the distillation process. In contrast, DomED strikes an exceptional balance between accuracy and calibration. We further compare DomED with more calibration strategies in Appendix H and show the reliability diagrams of ERM and DomED in Appendix I.

The analysis highlights two key aspects of its design. First, the raw ensemble of DomED teachers performs poorly on its own, confirming that our distillation process is essential to effectively integrate the specialists’ knowledge. Second,

Table 3. Uncertainty quantification performance on PACS. DomED achieves the best performance method among various single-inference models and ensemble distillation strategies.

Model	ERR↓	ECE↓	NLL↓	PRR↑
Ensemble	0.130	0.038	0.471	0.798
EoA	0.114	0.058	0.403	0.837
ERM	0.149	0.089	0.625	0.776
Temp. Scaling	0.147	0.048	0.487	0.773
MC Drop (p=0.5)	0.175	0.129	0.873	0.746
MC Drop (p=0.1)	0.157	0.112	0.755	0.769
CORAL	0.138	0.082	0.586	0.598
EnD ²	0.326	0.220	1.580	0.084
Scheme (a)	0.139	0.065	0.508	0.677
Scheme (b)	0.144	0.046	0.481	0.696
Scheme (c)	0.134	0.054	0.502	0.751
Scheme (d)	0.133	0.091	0.523	0.743
Scheme (e)	0.151	0.050	0.511	0.725
DomED (Teachers)	0.223	0.092	0.619	0.707
DomED (Ours)	0.131	0.044	0.473	0.787

Table 4. ROC-AUC of OOD detection. T.Unc and K.Unc refer to total uncertainty and knowledge uncertainty, respectively.

Models	PACS		OfficeHome	
	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.71	–	0.67	–
Ensemble	0.73	0.71	0.69	0.70
DomED	0.71	0.72	0.60	0.70

Models	VLCS		TerraInc.	
	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.51	–	0.75	–
Ensemble	0.53	0.54	0.78	0.79
DomED	0.57	0.69	0.77	0.75

DomED outperforms alternative distillation schemes (a)–(e) across all metrics, demonstrating the superiority of domain-specific teacher ensembling and complementary distillation. Overall, DomED achieves the best performance among all single-inference methods. While test-time ensembles serve as a strong performance ceiling, the competitive performance of DomED demonstrates the efficacy of our distillation strategy in transferring the benefits of an ensemble into a single efficient model. Finally, we present additional results on the impact of teacher diversity on model performance in Appendix C.

Out-of-Distribution Detection. We further assess the utility of DomED’s uncertainty estimates for out-of-distribution (OOD) detection. Following Malinin et al. (2019), we treat samples from the unseen test domain as OOD and samples from the source domains as in-distribution (ID). We use the area under the receiver operating characteristic curve (ROC-AUC) to measure the model’s ability to distinguish between these two groups using total uncertainty and knowledge un-

certainty. As shown in Table 4 (see Appendix L for more detailed results), DomED’s OOD detection performance is highly competitive with that of a full test-time ensemble (3 base models). On average, it closely matches the ensemble’s ability to identify novel, unseen domains, which is crucial for safe and reliable real-world deployment.

4. Related work

4.1. Domain Generalization

Domain Generalization (DG) addresses the performance degradation caused by distribution shifts (Moreno-Torres et al., 2012). Existing DG methods are often categorized into three main groups (Wang et al., 2022): data manipulation, representation learning, and learning strategies. Data manipulation methods augment training data to enhance diversity and quantity (Volpi et al., 2018; Zhou et al., 2021). Representation learning aims to find domain-invariant features (Du et al., 2021; Mahajan et al., 2021; Zhang et al., 2022; Dayal et al., 2023), with recent work exploring finer-grained control such as suppressing domain-sensitive channels (Guo et al., 2023) or establishing general objectives (Tan et al., 2024). Finally, learning strategies adapt techniques including ensemble learning (Arpit et al., 2022; Lee et al., 2022; Niu et al., 2023), meta-learning (Li et al., 2018a; Guan et al., 2023; Chen et al., 2023; Wang et al., 2025), and gradient operations (Shi et al., 2021; Ballas & Diou, 2025).

4.2. Ensemble Methods and Distillation

Ensemble methods have proven effective in improving OOD generalization. For instance, Arpit et al. (2022) employs an ensemble of moving average models to achieve better generalization performance but suffer from significant inference overhead. Efficient fusion techniques have been proposed to mitigate this, such as knowledge distillation (Hinton et al., 2015), weight averaging (Cha et al., 2021; Chu et al., 2022; Rame et al., 2022), and model merging (Ding et al., 2025). Specifically, distillation can be enhanced by gradient regularization (Wang et al., 2021) or language guidance (Huang et al., 2023). Beyond full-model ensembling, parameter-efficient aggregation has also gained attention, such as fusing task-specific adapters (Pfeiffer et al., 2021) or merging Low-Rank Adaptation (LoRA) modules (Hu et al., 2022; Yadav et al., 2023). In contrast to these parameter-space techniques that primarily focus on accuracy, DomED operates in the output space to explicitly transfer both generalization capabilities and epistemic uncertainty.

Theoretical work has shown that ensembles can improve accuracy by exploiting the “multi-view” structure of data (Allen-Zhu & Li, 2020). In the context of domain generalization, Zhou et al. (2021) introduce domain adap-

tive ensemble learning using shared feature extractors with domain-specific heads, a structure extended by Niu et al. (2023) via two-stage distillation. Subsequent works explore dynamic weighting based on domain relations during inference (Yao et al., 2023) or distilling experts from large pretrained models (Di Zhao et al., 2025). Unlike previous works that rely on general ensemble diversity, our work provides a specific theoretical grounding for complementary distillation, showing that it suppresses domain-specific biases in the distillation target, a property not guaranteed by standard ensemble averaging.

While Lee et al. (2022) propose self-distillation using logits with the same label, they ignore domain boundaries during teacher construction, resulting in a “single-view” teacher that lacks diversity. In contrast, DomED introduces a complementary strategy that explicitly leverages domain labels to construct diverse experts, which enables it to filter spurious features and provide calibrated uncertainty, capabilities that self-distillation methods inherently lack. Other approaches differ by relying on test-time statistics (Yan & Guo, 2025), positioning them closer to test-time adaptation, or by focusing on OOD detection (Wang et al., 2023a). However, a systematic study of how data allocation schemes can be tailored to the specific setting of domain generalization has been largely overlooked.

4.3. Uncertainty Quantification

In machine learning applications, distinguishing between data (aleatoric) uncertainty and model (epistemic) uncertainty is essential to understand the decisions or predictions made by models (Gal, 2016). Ensemble methods are particularly effective at separating and quantifying these two types of uncertainty (Lakshminarayanan et al., 2017). An intuitive approach to uncertainty quantification (Hüllermeier & Waegeman, 2021; Depeweg et al., 2017) derives total uncertainty as the entropy of the expected predictive distribution. This metric naturally decomposes into data uncertainty, defined as the expected entropy of individual predictions, and model uncertainty, representing the residual difference.

The Dirichlet distribution plays a crucial role in uncertainty quantification for classification models, particularly when distilling the knowledge of an ensemble into a single, efficient student model. As the conjugate prior of the categorical distribution, it provides a principled way to model a distribution over predicted probabilities. This approach is central to methods like Dirichlet prior networks (Tsiligkaridis, 2021; Malinin & Gales, 2018; Malinin et al., 2019) and evidential learning (Joo et al., 2020; Dawood et al., 2023; Schreck et al., 2024). However, training a model by directly minimizing the Dirichlet negative log-likelihood (NLL) can be problematic, as the loss function often degrades model accuracy in the process of improving uncertainty

estimates (Ryabinin et al., 2021).

5. Conclusion

In this work, we introduced DomED, a redesigned ensemble distillation framework for domain generalization that jointly addresses model accuracy and uncertainty quantification. By combining a theoretically grounded data allocation strategy that enhances teacher diversity and filters spurious correlations with a novel technique for decoupling mean and uncertainty distillation, DomED successfully transfers the knowledge of an ensemble into a single, efficient model. Our extensive experiments demonstrate that DomED achieves competitive accuracy while providing uncertainty estimates that rival a full, computationally expensive ensemble, underscoring the importance and feasibility of jointly optimizing for both objectives.

While our approach is more efficient than standard ensemble learning, its computational cost remains a limitation compared to single-model training, presenting a clear direction for future work on improving efficiency. Furthermore, the core principle of decoupled uncertainty distillation is broadly applicable. Extending this framework to other tasks, such as uncertainty-aware regression or dense prediction, is an exciting avenue for future research. Overall, this work provides a robust and practical foundation for building more reliable and uncertainty-aware models for domain generalization.

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Impact Statement

This paper aims to advance the field of Machine Learning by improving model robustness and uncertainty estimation across unseen domains. By enabling models to better quantify their limitations and generalize to out-of-distribution scenarios, our work contributes to the development of more reliable and safer machine learning deployments, particularly in high-stakes applications. We confirm that all experiments were conducted on publicly available academic datasets (e.g., PACS, OfficeHome) which contain no personally identifiable information, and our research involved no human subjects.

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A. Theoretical Analysis

We analyze the generalization capability of DomED by examining the properties of the distillation target. We demonstrate that while standard ERM exploits domain-specific spurious correlations to minimize training error, the complementary distillation strategy constructs a target that is conditionally independent of these spurious features, thereby guiding the student toward an invariant solution.

A.1. Setup and Assumptions

We adopt a linear structural equation model, a standard setting for analyzing the theoretical properties of domain generalization (Arjovsky et al., 2019; Krueger et al., 2021). We note that this linear setting serves as a theoretical justification to provide intuition for our design, rather than a formal guarantee for nonlinear models.

Data Generation Process. Let the input $\mathbf{x} \in \mathbb{R}^D$ be decomposed into invariant features $\mathbf{x}_{inv} \in \mathbb{R}^{d_c}$ and M blocks of domain-specific spurious features $\mathbf{x}_{sp} = [\mathbf{x}_{sp}^{(1)}, \dots, \mathbf{x}_{sp}^{(M)}]$, where $\mathbf{x}_{sp}^{(m)} \in \mathbb{R}^{d_s}$. The label $y \in \mathbb{R}$ is determined solely by the invariant features:

$$y = \mathbf{w}^* \cdot \mathbf{x}_{inv} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad (9)$$

where $\mathbf{w}^*$ represents the optimal invariant weights.

Domain Shift. In each source domain $m \in \{1, \dots, M\}$, the spurious features $\mathbf{x}_{sp}^{(m)}$ are spuriously correlated with the label, while others are noise. For a sample $(\mathbf{x}, y)$ drawn from domain distribution $\mathcal{D}_{tr}^m$:

$$\mathbf{x}_{sp}^{(k)} = \begin{cases} \mathbf{c} \cdot y + \xi_k, & \text{if } k = m \quad (\text{Spurious Correlation}) \\ \xi_k, & \text{if } k \neq m \quad (\text{Noise}) \end{cases}, \quad (10)$$

where $\mathbf{c} \in \mathbb{R}^{d_s}$ is a correlation vector and $\xi_k \sim \mathcal{N}(\mathbf{0}, \sigma_{sp}^2 \mathbf{I})$ is independent noise for all k . The goal is to learn a model that performs well on an unseen target domain $\mathcal{D}_{te}$ where spurious correlations are broken (i.e., $\mathbf{x}_{sp}^{(k)} \sim \mathcal{N}(\mathbf{0}, \sigma_{sp}^2 \mathbf{I})$ for all k). This modeling formalizes the intuition that spurious correlations are strictly domain-specific, and is corroborated by the observation that teachers trained on single domains often fail to generalize well to unseen domains (see Appendix C.1).

A.2. Spurious Feature Filtering

Let $\hat{\mathbf{w}}_S$ be the weights of a student model trained to minimize the squared error against the complementary distillation target $\bar{z}(\mathbf{x})$ on domain m .

Proposition A.1. *Under the data generation process defined above, the optimal student weights $\hat{\mathbf{w}}_S$ minimizing the complementary distillation loss satisfy:*

$$\hat{\mathbf{w}}_{S,sp}^{(m)} = \mathbf{0}.$$

That is, the student assigns zero weight to the spurious features of the training domain m , recovering the invariant solution (up to a scalar factor). In contrast, a model trained on hard labels y (ERM) assigns non-zero weight to $\mathbf{x}_{sp}^{(m)}$.

Proof.

Step 1: Teacher Convergence. Consider a teacher T^m trained on $\mathcal{D}_{tr}^m$ via least squares. The teacher learns weights $\hat{\mathbf{w}}_m = [\hat{\mathbf{w}}_{m,inv}, \hat{\mathbf{w}}_{m,sp}^{(1)}, \dots, \hat{\mathbf{w}}_{m,sp}^{(M)}]$. Since $\mathbf{x}_{sp}^{(m)}$ is correlated with y in $\mathcal{D}_{tr}^m$, the teacher exploits this feature to minimize training error. Standard linear regression results imply:

$$\hat{\mathbf{w}}_{m,inv} = \gamma \mathbf{w}^*, \quad \hat{\mathbf{w}}_{m,sp}^{(m)} = \lambda \mathbf{c}, \quad \hat{\mathbf{w}}_{m,sp}^{(k \neq m)} = \mathbf{0}, \quad (11)$$

where $\gamma, \lambda > 0$ are scaling factors depending on variances. Crucially, T^m learns a non-zero weight for feature m (the spurious shortcut) but zero weights for feature k (which is noise in domain m).

Step 2: Complementary Prediction. In DomED, to generate a target for an input $\mathbf{x}$ sampled from domain m ($\mathbf{x} \sim \mathcal{D}_{tr}^m$), we use the ensemble of complementary teachers $\{T^n\}_{n \neq m}$. The prediction of a single complementary teacher T^n on input $\mathbf{x}$ is:

$$\hat{y}_n(\mathbf{x}) = \hat{\mathbf{w}}_{n,inv} \cdot \mathbf{x}_{inv} + \sum_{k=1}^M \hat{\mathbf{w}}_{n,sp}^{(k)} \cdot \mathbf{x}_{sp}^{(k)}. \quad (12)$$

Substituting the teacher weights from Step 1:

$$\hat{y}_n(\mathbf{x}) = \gamma \mathbf{w}^* \cdot \mathbf{x}_{inv} + \lambda \mathbf{c} \cdot \mathbf{x}_{sp}^{(n)} + \sum_{k \neq n} \mathbf{0} \cdot \mathbf{x}_{sp}^{(k)}. \quad (13)$$

Note that the third term is zero because teacher T^n has zero weight for features $k \neq n$. The second term involves $\mathbf{x}_{sp}^{(n)}$. Since the input $\mathbf{x}$ comes from domain m (and $n \neq m$), $\mathbf{x}_{sp}^{(n)}$ is pure noise ξ_n . Thus, $\hat{y}_n(\mathbf{x}) = \gamma \mathbf{w}^* \cdot \mathbf{x}_{inv} + \text{noise}$.

Step 3: Distillation Target Aggregation. The distillation target $\bar{z}(\mathbf{x})$ is the average of $M - 1$ complementary teachers:

$$\bar{z}(\mathbf{x}) = \frac{1}{M-1} \sum_{n \neq m} \hat{y}_n(\mathbf{x}) = \gamma \mathbf{w}^* \cdot \mathbf{x}_{inv} + \underbrace{\frac{1}{M-1} \sum_{n \neq m} \lambda \mathbf{c} \cdot \xi_n}_{\epsilon_{ens}}. \quad (14)$$

As M increases, the noise term ϵ_{ens} averages to zero. Crucially, the spurious feature of the current domain, $\mathbf{x}_{sp}^{(m)}$, which is heavily present in the input $\mathbf{x}$, does not appear in the target $\bar{z}(\mathbf{x})$.

Step 4: Student Learning. The student minimizes $\mathcal{L} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{tr}^m} \|\mathbf{w}_S \cdot \mathbf{x} - \bar{z}(\mathbf{x})\|^2$. The optimal solution is the orthogonal projection of $\bar{z}(\mathbf{x})$ onto the span of the input features $\mathbf{x}$. Since $\bar{z}(\mathbf{x})$ is generated as a linear function of $\mathbf{x}_{inv}$ plus noise that is independent of $\mathbf{x}_{sp}^{(m)}$, the conditional expectation is:

$$\mathbb{E}[\bar{z}(\mathbf{x}) \mid \mathbf{x}_{inv}, \mathbf{x}_{sp}^{(m)}] = \gamma \mathbf{w}^* \cdot \mathbf{x}_{inv}.$$

Because $\mathbf{x}_{inv}$ is fully observed by the student, it is a sufficient statistic for the target $\bar{z}(\mathbf{x})$. The spurious feature $\mathbf{x}_{sp}^{(m)}$ provides no additional predictive power for $\bar{z}(\mathbf{x})$ (unlike for y , where it helps predict ϵ). Therefore, the regression coefficient for $\mathbf{x}_{sp}^{(m)}$ is zero: $\hat{\mathbf{w}}_{S,sp}^{(m)} = \mathbf{0}$. The student learns to rely exclusively on $\mathbf{x}_{inv}$. $\square$

A.3. Intuition

The core insight is that a teacher trained on Domain A is “blind” to the specific spurious shortcuts of Domain B . When we ask Teacher A to predict on an image from Domain B , it ignores the spurious features of B (treating them as noise) and relies primarily on the invariant features. Consequently, the ensemble of complementary teachers generates a “cleaned” supervision signal that retains the true semantics while filtering out the domain-specific biases present in the input image. The student, mimicking this cleaned signal, learns to ignore the spurious patterns inherent in its training data.

B. Computational Cost Analysis

B.1. Overhead of Training Teacher Models

A key practical advantage of DomED is its computational efficiency, despite using an ensemble of teacher models. This section demonstrates that the full training of teacher models is unnecessary for effective distillation, which significantly reduces the method’s computational cost. As shown in Figure 3, the distilled student models’ performance converges and stabilizes early in the training process (within approximately 1,000 steps), long before the teacher models have fully converged on their respective source domains. This observation indicates that while the guidance of teacher models is essential, the student’s performance is not sensitive to the teachers’ final, fully converged state.

This observation significantly reduces the computational cost of our method. For a training set with M domains, we can limit the training steps for each teacher to a fraction (e.g., $1/M$) of the steps used for a standard single-model baseline.¹

¹According to the DomainBed protocol, the number of training steps is 5,000 for PACS, OfficeHome, VLCS, TerraIncognita, and 15,000 for DomainNet.

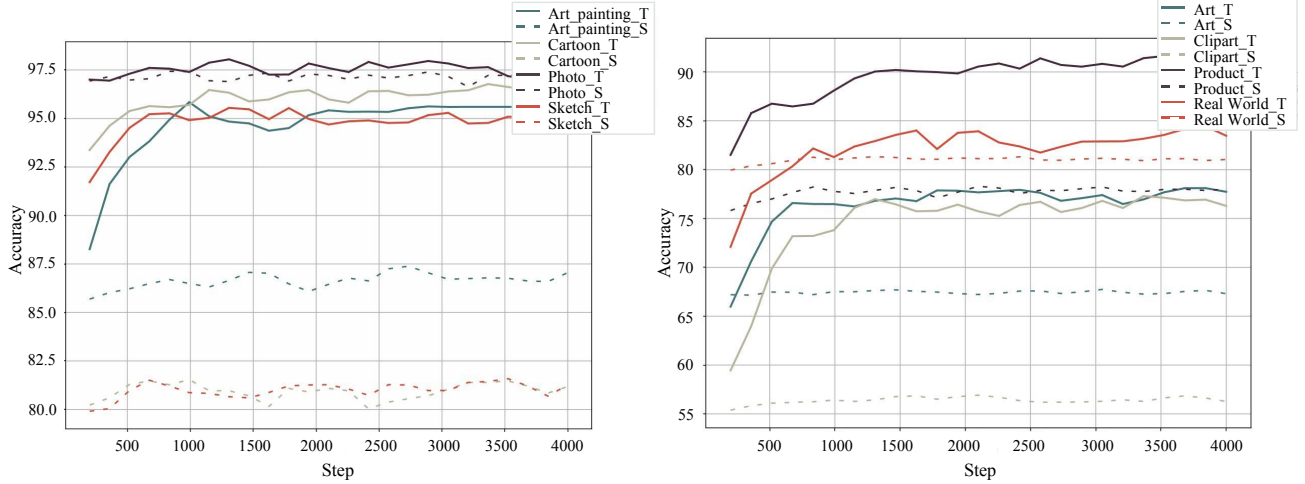


Figure 3. Impact of teacher training steps on student performance for (left) PACS and (right) OfficeHome. Solid lines represent teacher accuracy on their source domain, while dashed lines show the distilled student’s accuracy on the corresponding target domain. Note that student performance (dashed lines) converges and stabilizes early, long before the teacher models (solid lines) have fully converged.

Consequently, the total steps of training M teachers are comparable to training a single model on all domains, making DomED a computationally practical approach.

B.2. Training and Distillation Cost

We provide a detailed computational cost analysis on PACS in Table 5. The total training time for DomED is around 50% longer than ERM while achieving significantly better accuracy and calibration. The training of the teacher ensemble is particularly fast (0.78 h) and does not scale linearly with the number of domains. This efficiency stems from two factors: 1) Since each teacher sees only one domain, it is trained with a batch size $1/M$ of the standard ERM batch size, reducing the cost of each forward/backward pass. 2) As discussed in Appendix B.1, effective distillation does not require fully converged teachers because the student model saturates early, which allows us to significantly reduce the number of teacher training steps compared to a standard run.

Table 5. Computational cost and performance comparison on PACS (Target domain: Sketch).

Method	Wall-Time (h) ↓	Peak Mem (GB) ↓	Acc. (%) ↑	ECE ↓	NLL ↓
ERM	1.91	8.13	79.3	0.121	0.919
DNA (Chu et al., 2022)	3.23	8.17	79.8	0.087	0.708
DomED	2.86	8.13	81.6	0.028	0.609
↪ DomED (Teachers)	0.78	3.00	-	-	-
↪ DomED (Student)	2.08	8.13	-	-	-

When combined with SWAD (Cha et al., 2021), we apply weight averaging only during the distillation phase of the student model. As shown in Table 6, this incurs a small overhead compared to standard SWAD, yet achieves higher accuracy by combining the benefits of ensemble knowledge and flat-minima optimization (See Table 2).

Table 6. Total training time comparison with SWAD on PACS (Target: Sketch).

Method	Wall-Time (h) ↓
SWAD	2.41
DomED + SWAD	3.27

We further evaluate the training efficiency of DomED by comparing its wall-clock time against alternative data allocation

schemes. As summarized in Table 7, DomED is significantly faster than standard ensemble distillation (Scheme (b)) and schemes requiring multi-domain training (Schemes (d) and (e)). This efficiency is primarily due to our domain-specific teachers being trained on smaller $1/M$ data subsets, which provide effective supervision without requiring full convergence. Furthermore, DomED scales predictably to large-scale tasks. On DomainNet ($\sim 0.6M$ images), DomED requires a total of 13.04h (including 2.41h for teacher training and 10.63h for student distillation) compared to 9.62h for ERM.

Table 7. Wall-clock training time (hours) comparison of different ensemble and distillation strategies on the PACS dataset. DomED achieves the highest efficiency among explicit multi-model strategies.

Method	Teacher Training	Student Distillation	Total Time
Scheme (a)	1.91	1.89	3.80
Scheme (b)	2.34	2.26	4.60
Scheme (d) / (e)	1.56	2.04	3.60
Scheme (c)	0.78	2.14	2.92
DomED (Ours)	0.78	2.08	2.86

C. Analysis of Teacher Diversity

C.1. Visualizing Teacher Diversity

A central design principle of DomED is the use of a domain-specific data allocation scheme to foster the diversity required for effective ensemble learning. This section validates this strategy by visually analyzing the performance of teachers trained on distinct, single domains against those trained on dual-domain combinations. The accuracy heatmaps in Figure 4, Figure 5, Figure 6, and Figure 7 visualize this comparison across the PACS, OfficeHome, VLCS, and TerraIncognita datasets, respectively.

We observe a consistent pattern across these benchmarks where single-domain teachers (left panels) exhibit significantly higher specialization compared to their dual-domain counterparts (right panels). For instance, in the PACS dataset (Figure 4), a teacher trained exclusively on the “Art Painting” domain excels in its source domain but exhibits higher variance on others. In contrast, dual-domain teachers show more uniform performance across test domains. This empirical evidence supports our design choice: training each teacher on a single, distinct domain effectively encourages the development of specialized representations, which is a prerequisite for maximizing ensemble diversity.

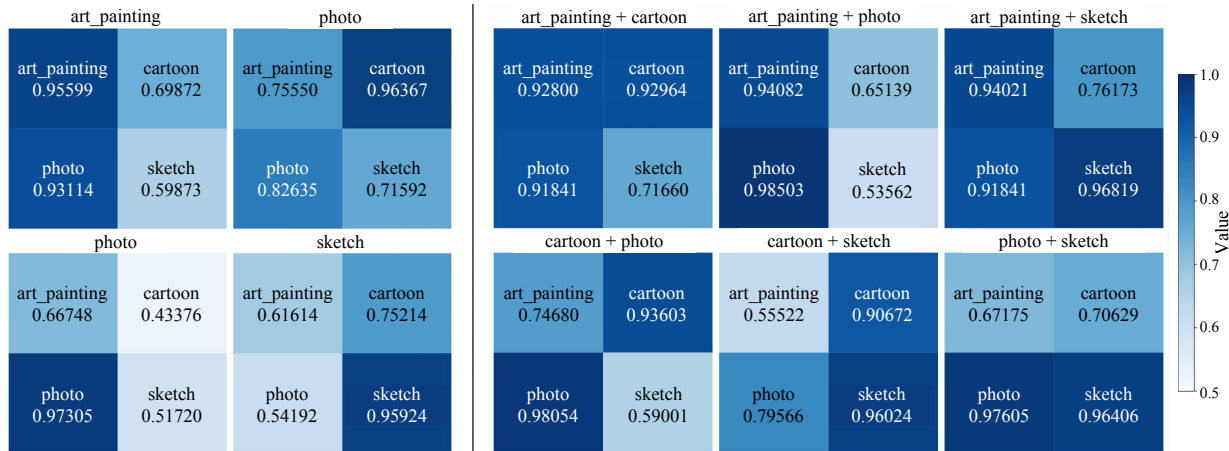


Figure 4. Accuracy heatmaps comparing single-domain (left) and dual-domain (right) teachers on the PACS dataset. Each block represents a trained teacher model, and each patch represents a test domain. The more varied color patterns on the left indicate higher specialization and diversity.

The heatmaps further reveal how diversity varies across datasets. On the OfficeHome dataset (Figure 5), single-domain teachers maintain reasonable accuracy on unseen domains. This indicates that despite the stylistic differences between

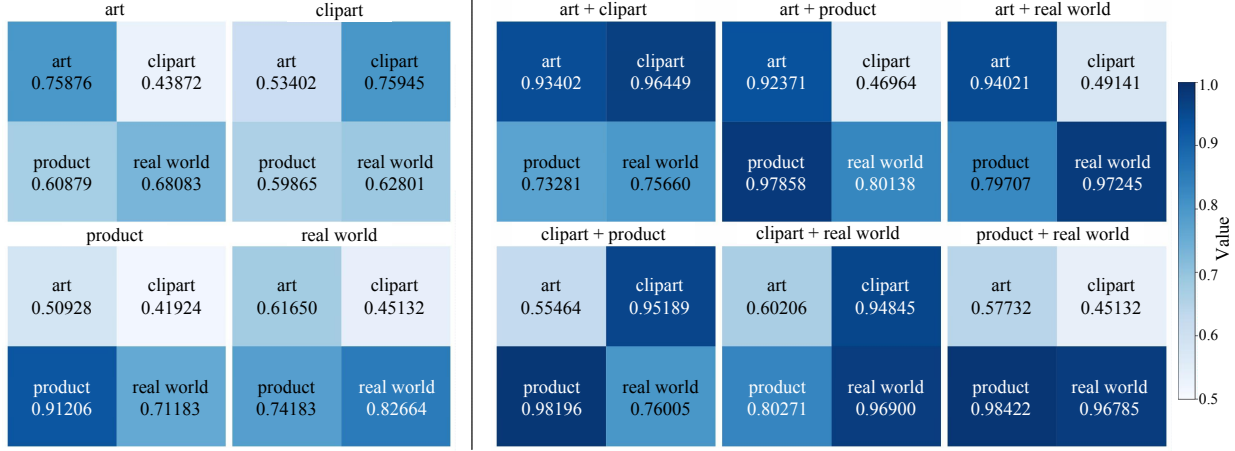


Figure 5. Accuracy heatmaps for single-domain (left) and dual-domain (right) teachers trained on the **OfficeHome** dataset.

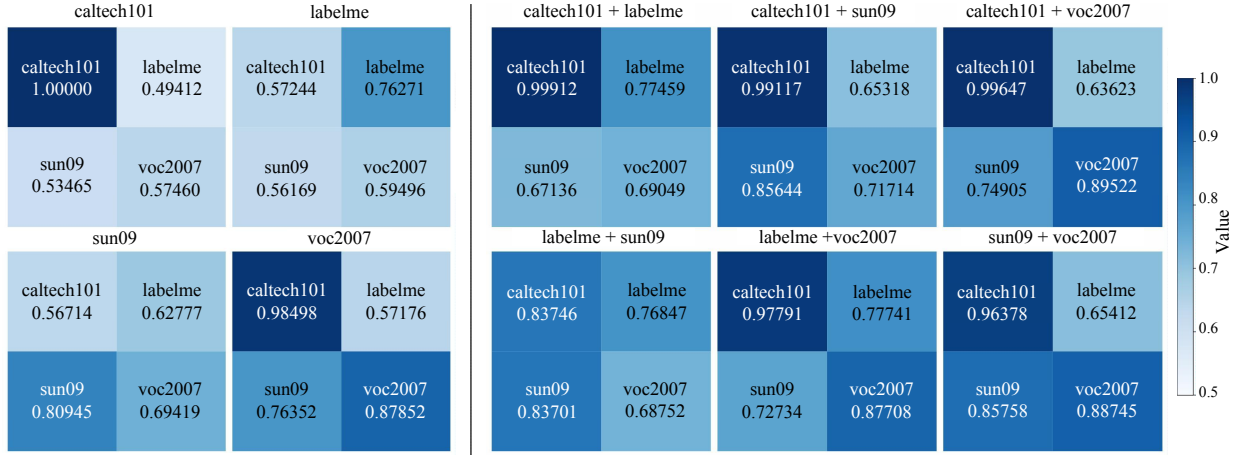


Figure 6. Accuracy heatmaps for single-domain (left) and dual-domain (right) teachers trained on the **VLCS** dataset.

domains (e.g., Art, Product), the teachers successfully learn transferable semantic features that enable effective cross-domain generalization. In contrast, we observe a different behavior on the TerraIncognita dataset. As shown in Figure 7, single-domain teachers on TerraIncognita exhibit a distinct pattern of high specialization, achieving high accuracy on their respective source domains but suffering significant performance drops on unseen domains. We hypothesize that this behavior stems from the nature of camera trap data, where static backgrounds (e.g., specific vegetation or terrain) encourage models to learn domain-specific shortcuts that do not transfer well. Despite this challenging scenario where teacher transferability is limited, DomED effectively recovers useful signals to outperform the ERM baseline (50.2% vs. 47.8%).

C.2. Impact of Teacher Diversity on Student Performance

This section quantifies the impact of teacher diversity on student performance through a controlled experiment on the PACS dataset. We compared our proposed single-domain allocation strategy (high granularity, high diversity) against a dual-domain allocation baseline (low granularity, low diversity) where teachers are trained on pairs of domains. As presented in Table 8, the results demonstrate that the higher diversity provided by single-domain teachers translates into superior student performance. While accuracy improves by 2.0%, the impact on uncertainty is most notable: the single-domain strategy reduces ECE by over 50% (from 0.107 to 0.044) compared to the dual-domain baseline.

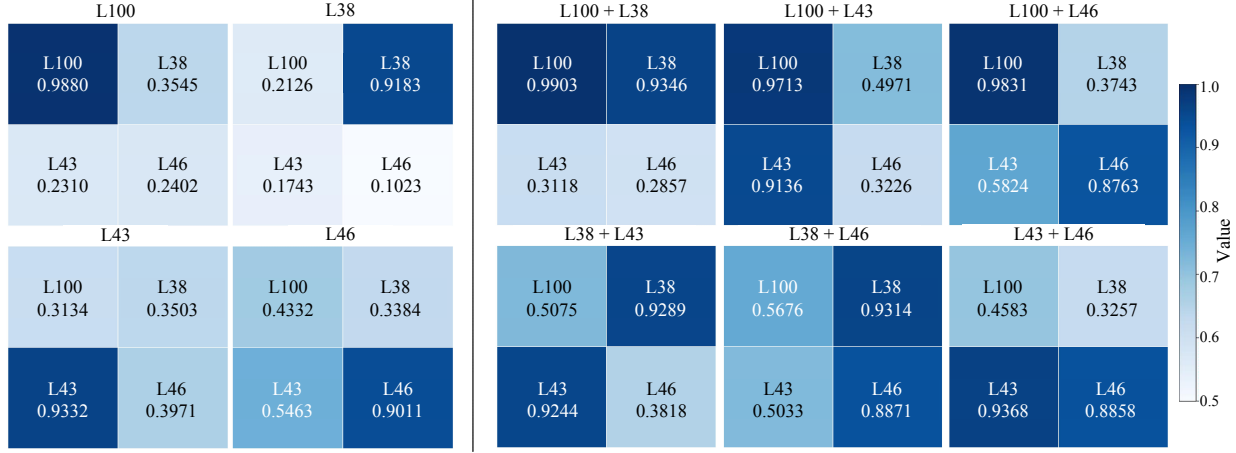


Figure 7. Accuracy heatmaps for single-domain (left) and dual-domain (right) teachers on the **TerraIncognita** dataset. Single-domain teachers exhibit high specialization.

Table 8. Impact of teacher diversity: Comparison between single-domain (DomED) and dual-domain allocation strategies on PACS.

Target Domain	Allocation	Diversity	Acc. (%)↑	ECE↓	NLL↓	PRR↑
Art Painting	Dual-domain	Low	85.1	0.109	0.521	0.772
	Single-domain (DomED)	High	87.5	0.049	0.497	0.781
Cartoon	Dual-domain	Low	78.3	0.076	0.726	0.654
	Single-domain (DomED)	High	81.5	0.038	0.628	0.738
Photo	Dual-domain	Low	96.0	0.094	0.200	0.882
	Single-domain (DomED)	High	97.1	0.061	0.158	0.936
Sketch	Dual-domain	Low	80.2	0.150	0.716	0.614
	Single-domain (DomED)	High	81.6	0.028	0.609	0.694
Average	Dual-domain	Low	84.9	0.107	0.541	0.730
	Single-domain (DomED)	High	86.9	0.044	0.473	0.787

C.3. Impact of Ensemble Size on Student Performance

Since our method ties the number of domain-specific teachers (or ensemble size) to that of training domains, varying the latter can affect teacher diversity as a result. To quantify the impact of ensemble size, we conducted an ablation study on the PACS dataset. We systematically train models on all possible 2-domain subsets ($M = 2$, each resulting in 2 domain-specific teachers) and compare their average performance against the standard 3-domain configuration ($M = 3$) as well as an ERM baseline trained on the same 2-domain subsets.

The results, summarized in Table 9, reveal that despite the reduction in teachers, DomED ($M = 2$) still outperforms the ERM baseline trained on the identical data subset. While accuracy is comparable, DomED maintains significantly better calibration (ECE 0.050 vs. 0.097). When compared to the full ensemble, reducing the size from 3 to 2 leads to a decrease in the Prediction Rejection Ratio ($0.787 \rightarrow 0.701$) and a slight increase in ECE. This confirms that the diversity provided by a larger set of domain experts is indeed a primary driver of high-quality uncertainty estimation. Crucially, the comparison indicates that while performance benefits from scaling up the number of domains, the method remains effective and robust when domain availability is limited.

Table 9. Impact of ensemble size (M) on student performance evaluated on PACS.

Metric	DomED ($M = 3$)	DomED ($M = 2$)	ERM ($M = 2$)
Accuracy $\uparrow$	86.9% ± 0.2	84.6% ± 0.3	84.3% ± 0.2
ECE $\downarrow$	0.044	0.050	0.097
NLL $\downarrow$	0.473	0.504	0.649
PRR $\uparrow$	0.787	0.701	0.649

D. Robustness to Weak Teachers

To evaluate the robustness of DomED against weak teachers trained on limited data, we performed a stress test on the PACS dataset, specifically using Sketch as the target domain. We simulated “amateur” teachers by reducing the training data of specific source domains (Art Painting, Photo, and Cartoon) to 50% and 25%. We compared our standard strategy (equal contribution) against a down-weighted strategy, where the weak teacher’s contribution to the soft target was manually suppressed (lowering the weight from 0.8 to 0.2).

Table 10. Robustness of different weighting strategies to data scarcity evaluated on PACS (Target domain: Sketch).

Data-Scarce Domain	Data %	Strategy	Student Acc (%) $\uparrow$	ECE $\downarrow$	NLL $\downarrow$
None (Baseline)	100%	Standard	81.6 ± 0.2	0.028	0.609
Art Painting	50%	Standard	81.4 ± 0.3	0.042	0.597
		Down-weighted	80.8 ± 0.2	0.049	0.614
	25%	Standard	80.5 ± 0.4	0.050	0.605
		Down-weighted	80.6 ± 0.3	0.061	0.626
Photo	50%	Standard	79.8 ± 0.3	0.054	0.618
		Down-weighted	80.2 ± 0.2	0.064	0.605
	25%	Standard	78.0 ± 0.5	0.041	0.686
		Down-weighted	76.5 ± 0.4	0.069	0.711
Cartoon	50%	Standard	77.8 ± 0.3	0.029	0.671
		Down-weighted	77.5 ± 0.2	0.027	0.703
	25%	Standard	78.6 ± 0.4	0.041	0.645
		Down-weighted	76.9 ± 0.3	0.035	0.738

As shown in Table 10, DomED proves to be remarkably resilient on the target domain. Even with only 25% of the training data, the performance drop is relatively small. Furthermore, manual down-weighting often degrades both accuracy and calibration (resulting in higher ECE). This suggests that even “amateur” teachers provide valuable diversity that aids generalization. The ensemble averaging mechanism, combined with the student’s hard-label loss anchor, naturally mitigates the noise from weaker teachers without requiring manual intervention.

E. Dark Knowledge Retention

The preservation of “dark knowledge”, the rich information contained in the non-target class probabilities, is essential for effective knowledge distillation (Hinton et al., 2015). However, this information is often lost when models become overly confident during training. We investigated this phenomenon by tracking the entropy of a model’s predictive distribution (i.e. a categorical distribution) on the PACS dataset throughout the training process. For the student model of DomED, we track the entropy of its mean predictive distribution.

In standard knowledge distillation settings, where teachers predict on their training data, confidence typically peaks early, leading to a rapid collapse in entropy (Figure 8(a) and (b)). However, the experiment reveals a fundamentally different behavior for DomED (Figure 8(c) and (d)). Since our complementary distillation strategy requires teachers to generate predictions for domains they have never seen during training, they are effectively performing OOD inference. Consequently, they naturally yield “softer” probability distributions with stable, elevated entropy (≈ 0.49 nats on average) throughout the

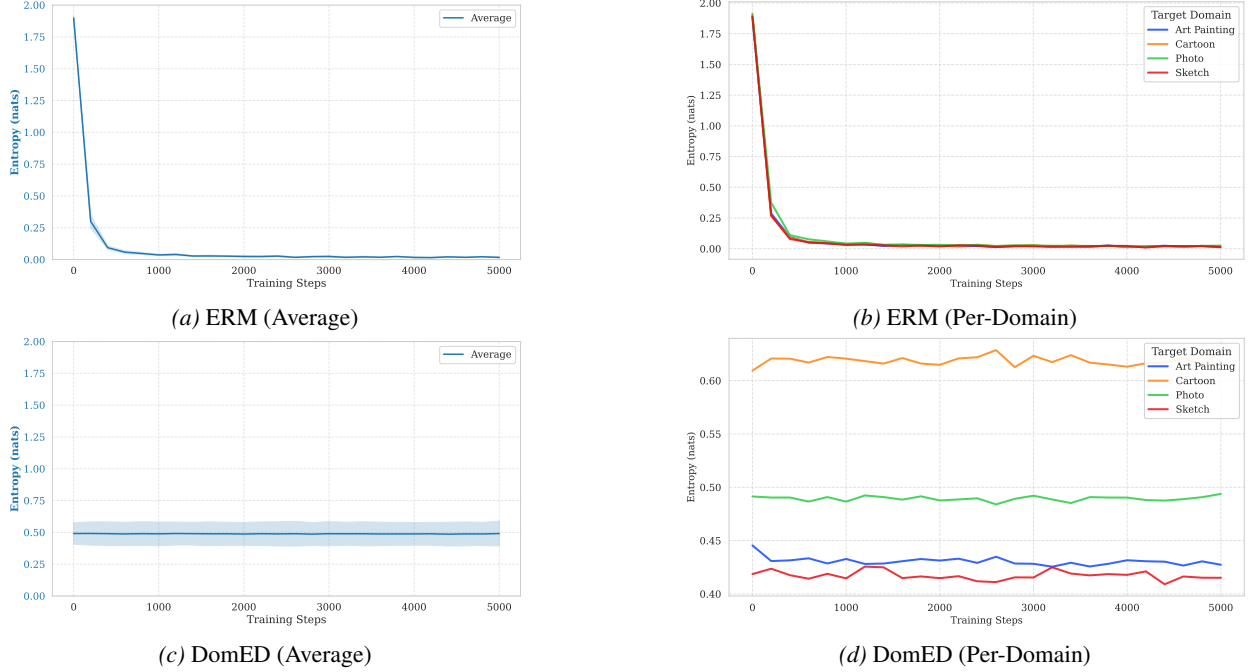


Figure 8. Entropy of the predictive distribution during model training. The average entropy (a) and per-domain entropy (b) of ERM rapidly collapse during training. In contrast, the average entropy (c) and per-domain entropy (d) of the DomED student remains stable at a high level throughout training (distillation).

entire training process.

To further validate whether the retained dark knowledge is sufficient, we evaluated an intervention strategy where we doubled the distillation temperature during the late training phase. We found that this yielded no notable improvement in accuracy or calibration. This leads us to conclude that the natural “OOD signal” provided by the complementary allocation is already sufficient to preserve dark knowledge, rendering additional entropy-preserving constraints unnecessary in our framework.

F. Ablation on Loss Components

We conduct an ablation study on the components of our proposed loss function, $\mathcal{L}'_S = \mathcal{L}_S + \beta\mathcal{L}_{\text{Dir}}$, to validate our decoupled distillation approach. This analysis compares the performance when using only the standard distillation loss ($\mathcal{L}_S$), only the Dirichlet NLL loss ($\mathcal{L}_{\text{Dir}}$), and our complete decoupled loss. The results are presented in Table 11. Relying solely on $\mathcal{L}_{\text{Dir}}$ leads to a significant degradation in classification accuracy, which is consistent with the issue of penalizing confident predictions. While $\mathcal{L}_S$ alone provides a strong performance baseline, our complete loss, $\mathcal{L}_S + \mathcal{L}_{\text{Dir}}$, consistently maintains or even slightly improves accuracy across multiple benchmarks.

To verify whether this design effectively resolves the optimization conflict discussed in Section 2.3 (i.e., balancing mean prediction and uncertainty spread), we further evaluate accuracy and uncertainty metrics on the PACS dataset. As shown in Table 12, combining the two losses significantly improves both ECE and PRR compared to using $\mathcal{L}_{\text{Dir}}$ alone. These empirical results confirm that $\mathcal{L}_S$ is essential for anchoring the mean prediction, which in turn allows $\mathcal{L}_{\text{Dir}}$ to focus primarily on shaping the distribution’s spread without compromising convergence.

G. Sensitivity to Hyperparameters

G.1. Soft Label Weight and Temperature

We investigate the impact of the soft label weight (λ) and the temperature (τ) on model accuracy. Figure 9 and Figure 10 present heatmaps of model performance on the PACS and VLCS datasets, respectively. The analysis shows that performance varies smoothly with these hyperparameters. For instance, on the PACS dataset, the Photo domain is least sensitive, while

Table 11. Ablation study of the distillation loss components on three benchmarks, reporting classification accuracy (%).

<i>PACS</i>					
Loss Component	Art Painting	Cartoon	Photo	Sketch	Avg.
Only $\mathcal{L}_{\text{Dir}}$	64.49	60.78	87.20	59.07	67.88
Only $\mathcal{L}_S$	87.46	81.48	97.08	81.55	86.89
$\mathcal{L}_{\text{Dir}} + \mathcal{L}_S$	87.52	81.51	97.10	81.64	86.94
<i>VLCS</i>					
Loss Component	Caltech	LabelMe	SUN09	VOC2007	Avg.
Only $\mathcal{L}_{\text{Dir}}$	76.15	57.69	75.36	69.60	69.70
Only $\mathcal{L}_S$	98.26	65.08	78.23	78.21	79.94
$\mathcal{L}_{\text{Dir}} + \mathcal{L}_S$	98.51	65.31	78.42	78.40	80.14
<i>TerraIncognita</i>					
Loss Component	Loc. 100	Loc. 38	Loc. 43	Loc. 46	Avg.
Only $\mathcal{L}_{\text{Dir}}$	28.61	47.27	20.34	27.56	30.90
Only $\mathcal{L}_S$	55.12	46.47	57.81	38.53	49.48
$\mathcal{L}_{\text{Dir}} + \mathcal{L}_S$	55.31	47.55	59.76	35.55	49.54

Table 12. Impact of loss components on accuracy and uncertainty metrics evaluated on PACS.

Loss Component	Acc. (%) $\uparrow$	ECE $\downarrow$	NLL $\downarrow$	PRR $\uparrow$	K.Unc $\uparrow$
Only $\mathcal{L}_{\text{Dir}}$	67.88	0.397	0.917	0.640	0.672
Only $\mathcal{L}_S$	86.89	0.080	0.542	0.771	0.719
$\mathcal{L}_{\text{Dir}} + \mathcal{L}_S$	86.94	0.044	0.473	0.787	0.723

the Cartoon domain exhibits low sensitivity to changes in τ but high sensitivity to changes in λ . Overall, these results indicate that DomED’s performance is robust across a reasonable range of hyperparameter settings, validating the stability of our approach.

G.2. Weight of the Dirichlet NLL Loss

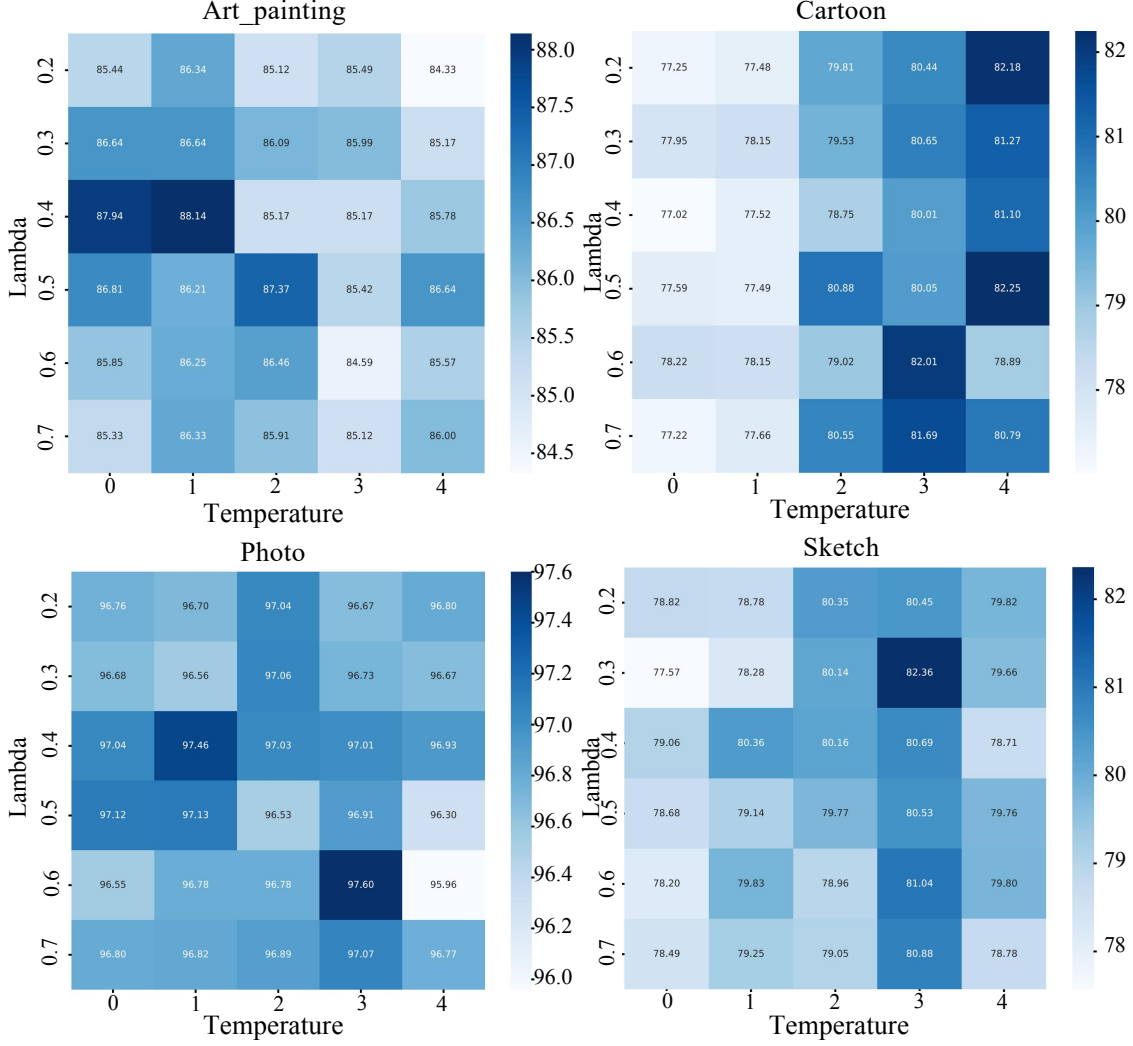
The hyperparameter β in our decoupled distillation loss, $\mathcal{L}'_S = \mathcal{L}_S + \beta\mathcal{L}_{\text{Dir}}$, is crucial for balancing accuracy and uncertainty transfer. To analyze our method’s robustness to this hyperparameter, we investigate its impact on both the loss components and the final uncertainty estimates. As shown in Figure 11, we first examine the trade-off between the two loss components. Increasing β places more weight on the uncertainty-focused $\mathcal{L}_{\text{Dir}}$, causing its value to decrease at the expense of the accuracy-focused $\mathcal{L}_S$. The results indicate that a small value, such as $\beta = 0.01$, strikes an effective balance, allowing $\mathcal{L}_{\text{Dir}}$ to converge sufficiently without significantly degrading the performance of $\mathcal{L}_S$.

This balance then translates directly to the final uncertainty estimates, as confirmed in Figure 12. The same region around $\beta = 0.01$ successfully preserves a high degree of both total and knowledge uncertainty from the ensemble. These results validate our core claim: the principled decoupling, controlled by a small β , effectively distills an ensemble’s uncertainty without the accuracy trade-off typically associated with relying solely on the Dirichlet NLL objective.

H. Comparison with Alternative Calibration Strategies

We evaluated DomED against Post-hoc Temperature Scaling (TS) (Guo et al., 2017), Label Smoothing (LS) (Szegedy et al., 2016), and Focal Loss (Lin et al., 2017) on the PACS dataset. The results are summarized in Table 13.

Post-hoc TS reduces ECE but yields negligible gains in error detection (PRR) or OOD detection. While Label Smoothing achieves a high OOD-AUC, it incurs a performance trade-off, resulting in a 2.7% increase in classification error and a significantly higher ECE compared to DomED. Focal Loss similarly results in a higher error rate and worse calibration.

Figure 9. Test accuracies on PACS as λ and τ vary.Table 13. Comparison of DomED with alternative calibration strategies on PACS. We evaluate Post-hoc Temperature Scaling (TS), Label Smoothing (LS, $\epsilon = 0.1$), and Focal Loss ($\gamma = 2.0$) within the same complementary allocation regime.

Category	Method / Configuration	ERR ↓	ECE ↓	NLL ↓	PRR ↑	OOD-AUC ↑
Baseline	$\mathcal{L}_S$ only	0.1311	0.080	0.542	0.771	71.96
	+ Post-hoc TS (Guo et al., 2017)	0.1311	0.065	0.510	0.773	71.96
Alternatives	Label Smoothing (Szegedy et al., 2016)	0.1582	0.113	0.550	0.669	73.17
	Focal Loss (Lin et al., 2017)	0.1704	0.129	0.590	0.582	72.20
Ours	DomED ($\mathcal{L}_S + \mathcal{L}_{Dir}$)	0.1306	0.044	0.473	0.787	72.25
	+ Post-hoc TS	0.1306	0.105	0.580	0.785	72.26

Furthermore, we investigate whether post-hoc temperature scaling (TS) can improve state-of-the-art DG methods (e.g., MIRO and SWAD). As shown in Table 14, while TS can reduce ECE for over-confident models, it is strictly a monotonic transformation. Consequently, it cannot alter relative confidence rankings, leaving true epistemic capabilities such as PRR and OOD-AUC entirely unchanged across these methods.

In contrast, DomED simultaneously achieves a highly competitive error rate (0.1306), the lowest ECE (0.044), and the

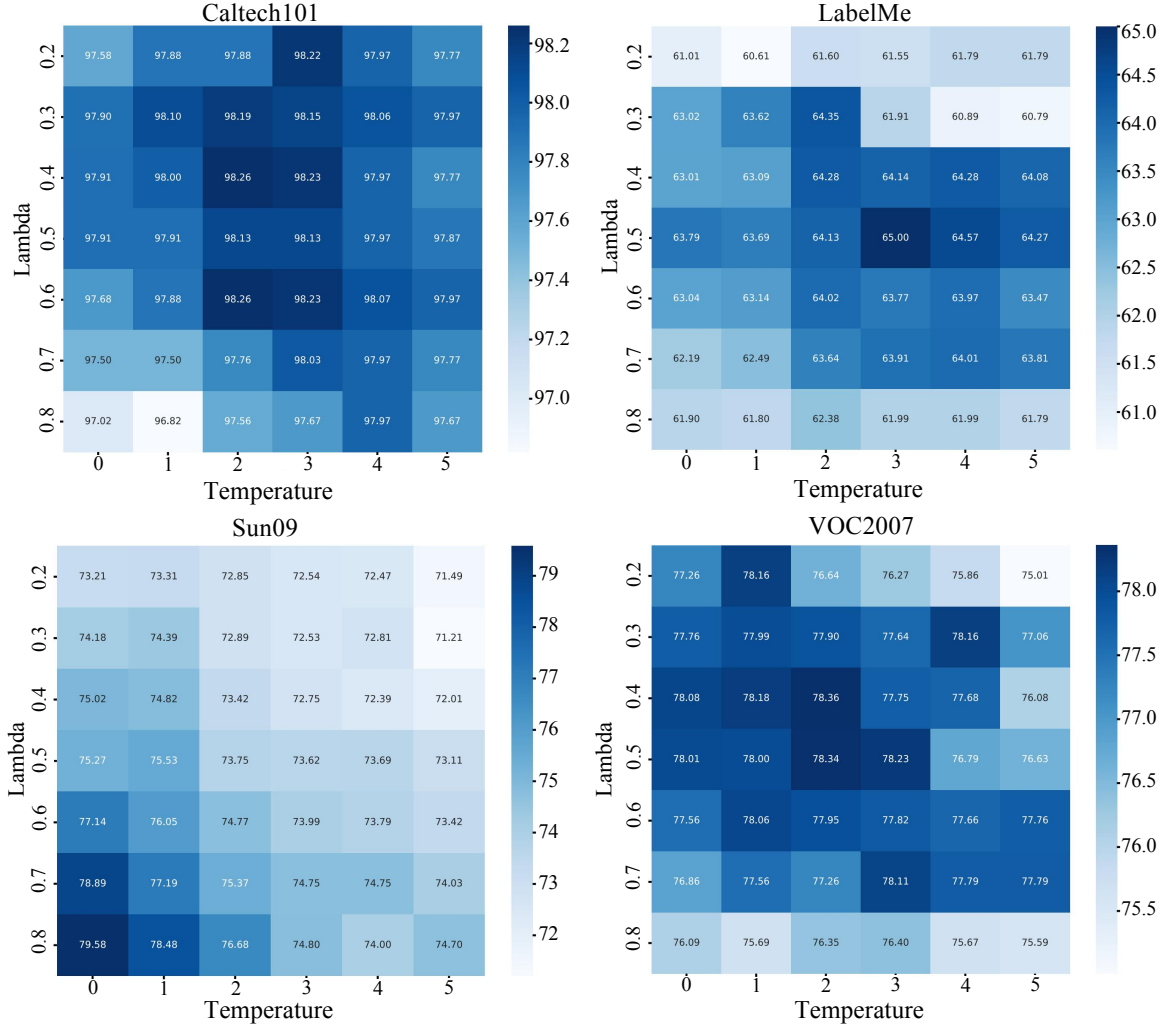
Figure 10. Test accuracies on VLCS as λ and τ vary.

Table 14. Uncertainty quantification performance with post-hoc temperature scaling (TS) on the PACS dataset.

Method (ResNet-50)	ERR ↓	ECE ↓	NLL ↓	PRR ↑	OOD-AUC ↑
ERM	0.149	0.089	0.625	0.776	71.21
ERM + TS	0.149	0.048	0.487	0.773	71.22
MIRO	0.146	0.082	0.615	0.718	71.75
MIRO + TS	0.146	0.055	0.585	0.715	71.75
SWAD	0.119	0.076	0.598	0.784	68.78
SWAD + TS	0.119	0.052	0.465	0.786	68.76
DomED (Ours)	0.131	0.044	0.473	0.787	72.25
DomED + TS	0.131	0.105	0.580	0.785	72.26

highest PRR (0.787). Notably, applying post-hoc TS to DomED degrades calibration, suggesting that our method learns intrinsically calibrated probabilities. We attribute this strong performance partly to the fact that the Dirichlet distribution is the conjugate prior of the categorical distribution. This makes it a theoretically natural choice for capturing the uncertainty inherent in an ensemble.

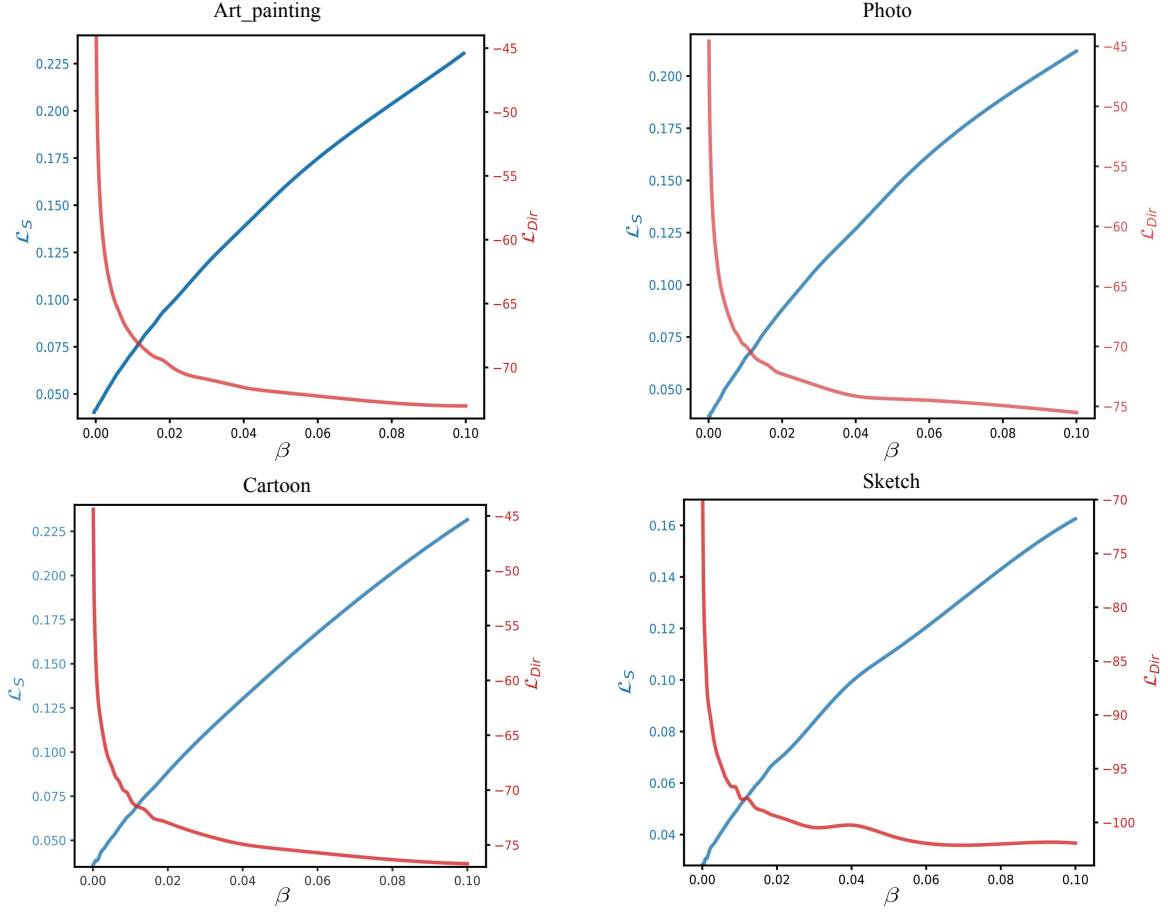


Figure 11. Converged values of the loss components $\mathcal{L}_S$ (blue) and $\mathcal{L}_{Dir}$ (red) as a function of β . The plot illustrates the trade-off between optimizing for accuracy ($\mathcal{L}_S$) and uncertainty ($\mathcal{L}_{Dir}$) on the PACS dataset.

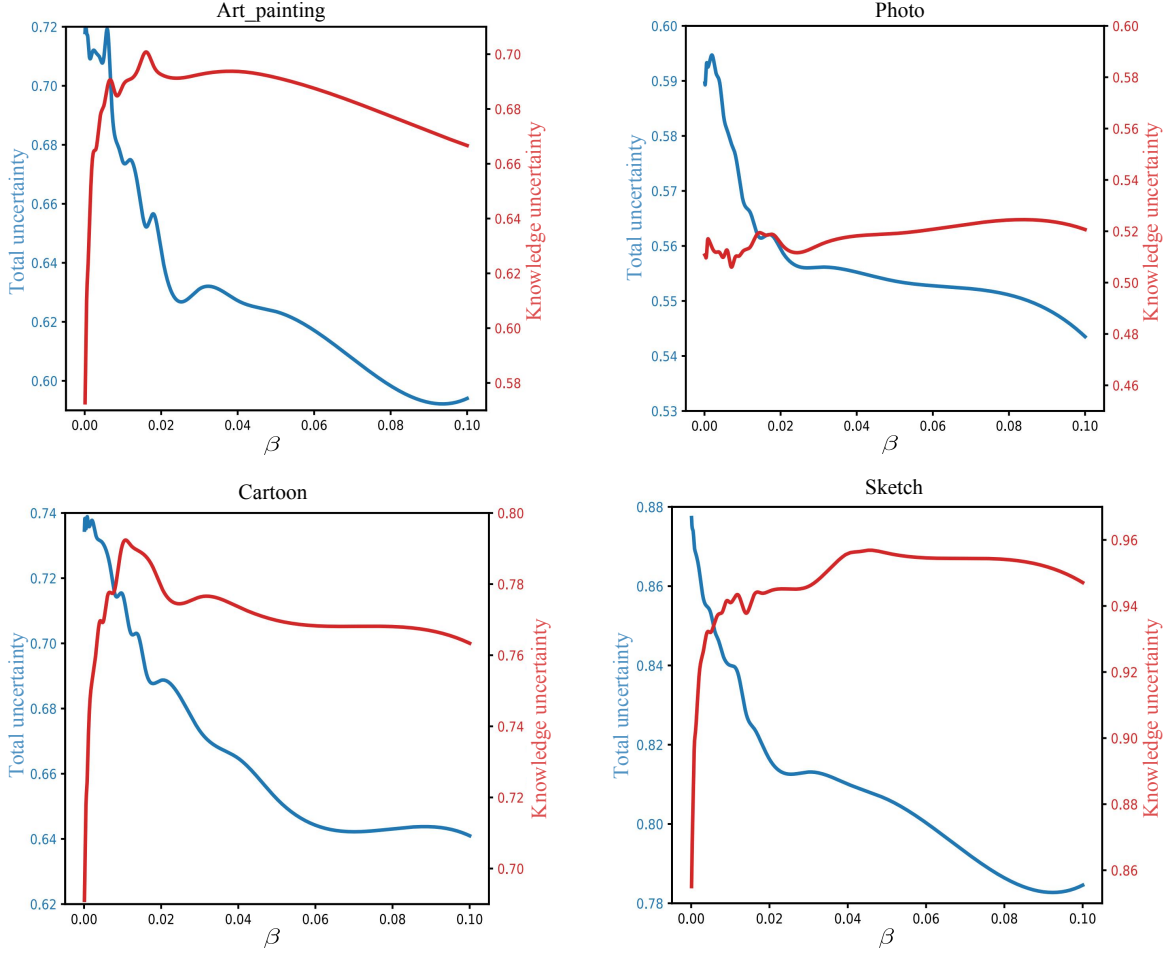


Figure 12. Impact of β on the final uncertainty estimates for the PACS dataset. The plots show that a small β (e.g., 0.01) is sufficient to distill a high degree of both total and knowledge uncertainty from the teacher ensemble.

I. Reliability Diagrams

We further corroborate the robust calibration of DomED with reliability diagrams, which visualize the alignment between predictive confidence and empirical accuracy. As illustrated in Figures 13 and 14, the ERM baseline exhibits well-calibrated predictions on IID data, but significantly deteriorates on OOD data. Conversely, DomED preserves a higher predictive entropy during training, which acts as an intrinsic regularization mechanism to prevent over-fitting to source domains. As a result, DomED effectively mitigates overconfidence and maintains reliable calibration across both IID and OOD scenarios.

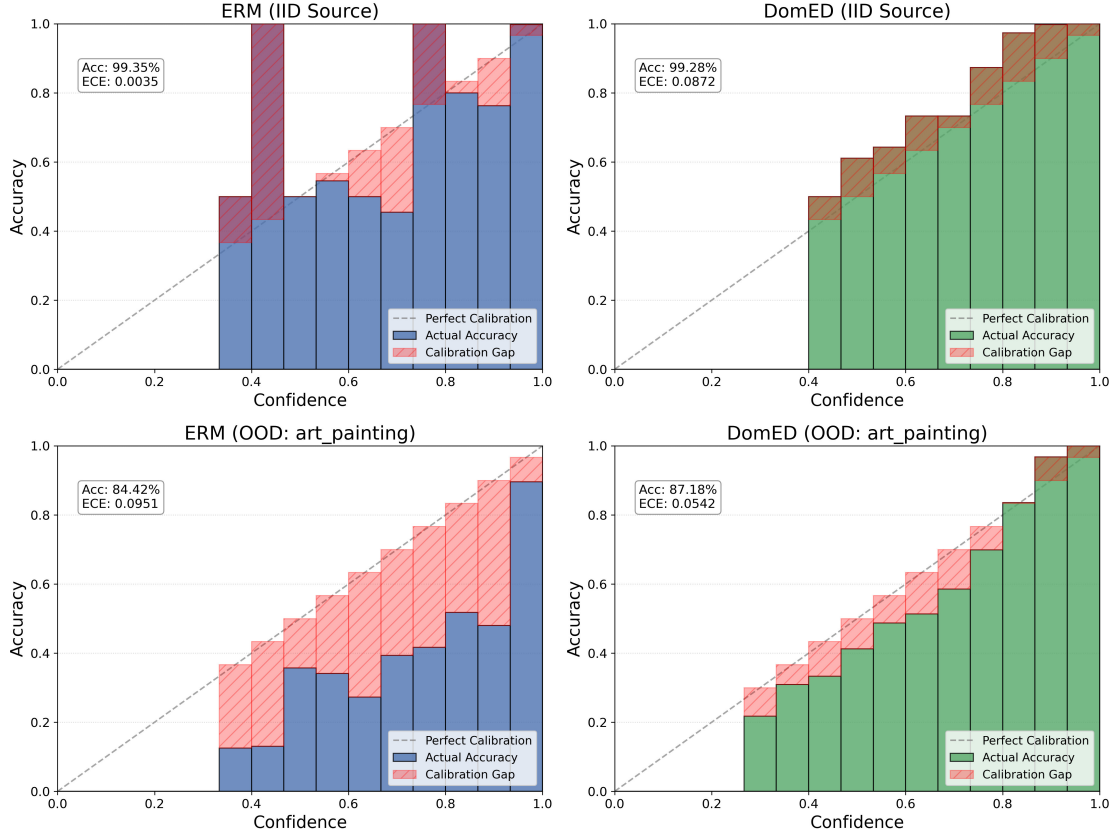


Figure 13. Reliability diagrams on the Art Painting domain of PACS.

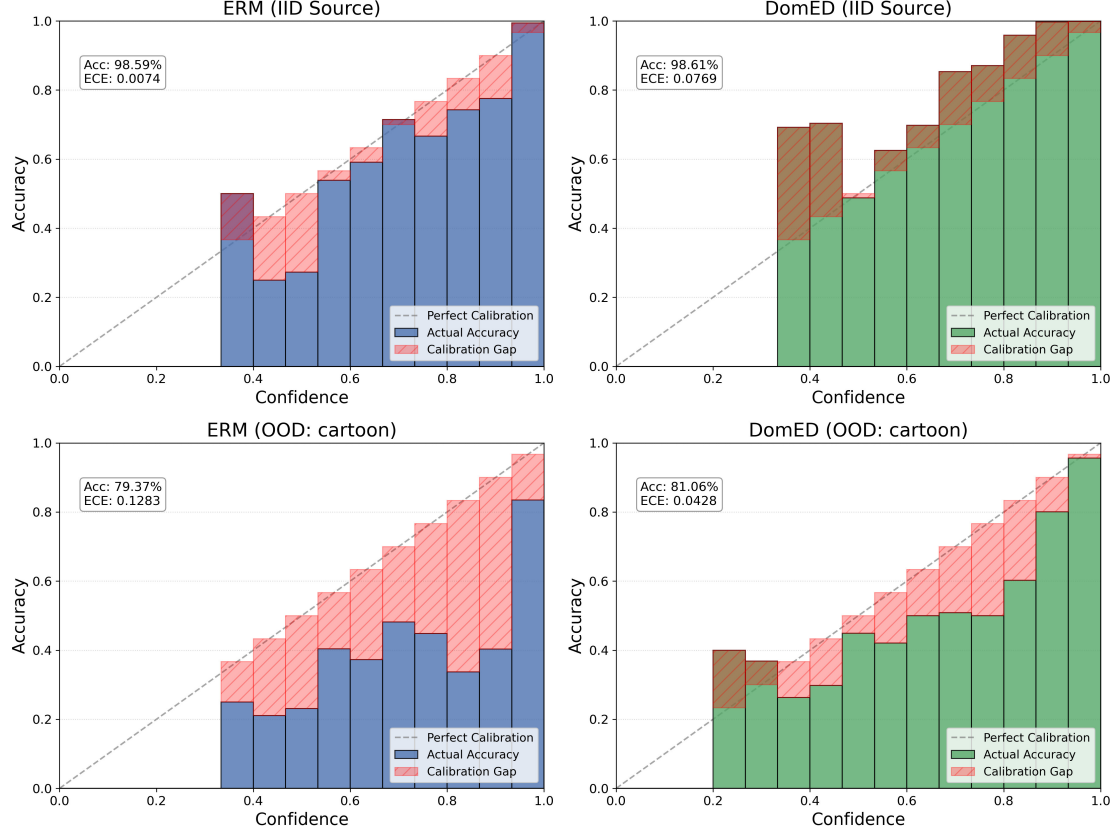
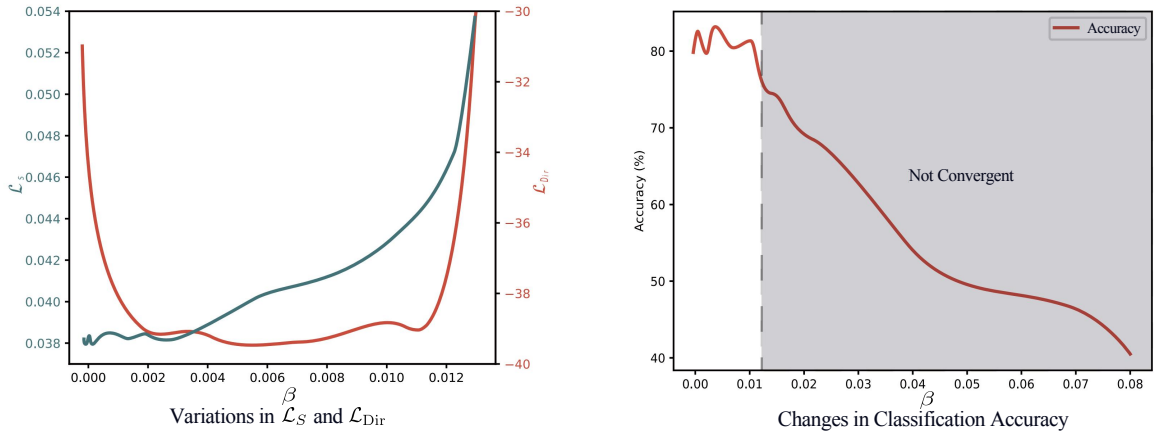
J. Analysis of an Alternative Decoupling Strategy

The standard Dirichlet NLL loss ($\mathcal{L}_{\text{Dir}}$) can have a large gradient norm compared to the cross-entropy loss, which can destabilize training and harm classification accuracy. In the main paper, we address this with a simple and effective decoupled loss, $\mathcal{L}'_S = \mathcal{L}_S + \beta \mathcal{L}_{\text{Dir}}$. Here, we analyze a more complex alternative to provide further justification for our chosen design. The alternative strategy aims to directly isolate the learning of the mean prediction from the uncertainty. The mean of the Dirichlet distribution is given by $\mathbb{E}[\pi_c] = \alpha_c / \alpha_0 = e^{z_c} / \sum_k e^{z_k}$, which has the same form as the standard softmax function. The uncertainty, however, is controlled by the precision α_0 , which is determined by the sum of logits, $z_0 = \sum_c z_c$. This implies that z_0 introduces an extra degree of freedom for learning uncertainty. Therefore, we propose decoupling the uncertainty distillation by using Equation (7) to learn z_0 and Equation (5) to learn $\mathbb{E}[\pi_c]$. This ensures that the mean categorical distribution is learned regularly without compromising model accuracy. To achieve this, we reparameterize the z_c 's in Equation (7) as follows:

$$z_c = \text{stop_gradient}(z_c - z_0) + z_0, \quad (15)$$

where $\text{stop_gradient}(\cdot)$ operation blocks gradients from directly flowing to z_c while allowing them to pass through z_0 .

However, our experiments showed this approach to be highly unstable. As illustrated in Figure 15, this reparameterization causes the Dirichlet NLL to diverge for all but a very narrow range of β . When β exceeds a small threshold (approximately 0.012), the model fails to converge, leading to a collapse in classification accuracy. This investigation highlights the difficulty


 Figure 14. Reliability diagrams on the *Cartoon* domain of PACS.

 Figure 15. The alternative decoupling strategy using ‘stop_gradient’ is highly unstable. (Left) Both loss components, $\mathcal{L}_S$ and $\mathcal{L}_{Dir}$, diverge with increasing β on the PACS dataset. (Right) Classification accuracy collapses when β exceeds a small threshold, indicating a failure to converge.

of directly manipulating the gradients of the Dirichlet NLL via explicit reparameterization. It motivated our decision to use the simpler and far more robust decoupled loss function presented in the main paper, which effectively balances the two objectives without requiring explicit gradient surgery and demonstrates stable performance across all benchmarks.

K. Robustness to Model Architecture

Table 15. Performance on PACS across diverse model architectures (Architecture), including ResNet-18 and the Vision Transformer Base model (ViT-B/16). DomED consistently improves accuracy and uncertainty estimation over the ERM baselines, demonstrating its robustness.

Architecture	Model	ERR↓	ECE↓	NLL↓	PRR↑	T.Unc	K.Unc
ResNet-18	ERM	0.192	0.128	0.895	0.775	0.723	-
	Ensemble	0.172	0.066	0.697	0.772	0.724	0.698
	DomED	0.175	0.043	0.579	0.745	0.713	0.729
ResNet-50	ERM	0.149	0.089	0.625	0.776	0.714	-
	Ensemble	0.130	0.038	0.471	0.798	0.730	0.711
	DomED	0.131	0.044	0.473	0.787	0.705	0.723
ViT-B/16	ERM	0.211	0.155	0.988	0.688	0.723	-
	Ensemble	0.203	0.050	0.722	0.711	0.724	0.721
	DomED	0.146	0.079	0.534	0.785	0.623	0.692

We evaluate the architectural robustness of DomED by repeating our experiments on the PACS dataset with ResNet-18 (He et al., 2016) and a vision transformer (ViT-B/16) (Dosovitskiy et al., 2021). As shown in Table 15, the results confirm that DomED is a broadly applicable framework that consistently improves upon the ERM baselines. With the smaller ResNet-18, DomED not only improves accuracy but also yields superior uncertainty estimates (ECE and NLL) compared to the full test-time ensemble, demonstrating its effectiveness on more compact models. The results on the ViT-B/16 are particularly compelling. Vision transformers are known to struggle with generalization on smaller DG datasets, which is reflected in the high error rates of the ERM and ensemble baselines. In this challenging scenario, DomED provides a remarkable improvement, reducing the classification error by approximately 6 percentage points compared to the ensemble. This suggests that our distillation strategy may function as a powerful regularizer, validating DomED as a robust framework for diverse model families in the domain generalization setting.

Table 16. Domain generalization performance (%) evaluated on the ResNet-18 backbone. “-” indicates results not reported.

Method	PACS	OfficeHome	VLCS	TerraIncognita	Avg.
ERM	83.0	65.7	77.2	41.4	66.8
ARM (Zhang et al., 2021)	81.7	64.4	76.6	42.6	66.3
XDED (Lee et al., 2022)	83.8	65.0	74.8	42.5	66.5
BOLD (Di Zhao et al., 2025)	82.0	66.2	-	-	-
DomED (Ours)	82.5	66.3	77.8	42.7	67.3

We further extend the evaluation on ResNet-18 to more datasets (PACS, OfficeHome, VLCS, and TerraIncognita), and compare the generalization performance against ERM and other methods that have also (or only) been evaluated on ResNet-18. As shown in Table 16, DomED consistently demonstrates highly competitive performance, achieving the best average accuracy.

L. Additional Results on Uncertainty Quantification

This section provides a detailed breakdown of the uncertainty quantification results summarized in Section 3.3. We present the performance metrics for each leave-one-domain-out split, supplementing the averaged results shown in the main paper. Table 17 contains the per-domain calibration and reliability metrics corresponding to Table 3. Table 18 provides the per-domain OOD detection performance corresponding to Table 4.

Table 17. Detailed uncertainty quantification results on the PACS dataset. The table is split into two parts for better readability: Art Painting and Cartoon (top), Photo and Sketch (bottom). The best and second-best results are indicated by **bold** and underlined, respectively.

Model	Target: Art Painting				Target: Cartoon			
	ERR↓	ECE↓	NLL↓	PRR↑	ERR↓	ECE↓	NLL↓	PRR↑
Ensemble	0.139	0.026	0.489	0.830	0.160	0.057	0.633	0.761
EoA	0.111	0.048	0.375	0.894	0.143	0.099	0.627	0.774
ERM	0.152	0.065	0.496	0.808	0.200	0.125	0.821	0.731
Temp. Scaling	0.164	0.069	0.576	0.759	0.191	0.061	0.599	0.758
MC Drop (p=0.5)	0.163	0.100	0.686	0.772	0.238	0.189	1.414	0.678
MC Drop (p=0.1)	0.175	0.109	0.736	0.766	0.203	0.165	1.241	0.703
CORAL	0.139	0.071	0.514	0.670	<u>0.181</u>	0.117	0.754	0.484
EnD ²	0.274	0.131	1.262	0.142	0.481	0.358	2.590	0.007
Scheme (a)	<u>0.127</u>	0.063	0.466	0.702	0.188	0.062	0.688	0.722
Scheme (b)	0.132	<u>0.041</u>	<u>0.435</u>	0.771	0.190	0.062	0.658	0.683
Scheme (c)	0.128	0.060	0.490	<u>0.795</u>	0.190	0.042	0.698	0.679
Scheme (d)	0.125	0.125	0.477	0.748	0.179	0.053	0.697	0.710
Scheme (e)	0.133	0.037	0.481	0.726	0.189	<u>0.040</u>	0.584	0.688
DomED (Teachers)	0.212	0.057	0.407	0.711	0.276	0.125	0.802	0.690
DomED (Ours)	0.125	0.049	0.497	0.781	0.185	0.038	0.628	<u>0.738</u>

Model	Target: Photo				Target: Sketch			
	ERR↓	ECE↓	NLL↓	PRR↑	ERR↓	ECE↓	NLL↓	PRR↑
Ensemble	0.038	0.012	0.125	0.945	0.184	0.057	0.637	0.657
EoA	0.019	0.011	0.049	0.958	0.166	0.076	0.561	0.722
ERM	0.038	0.045	0.265	0.891	0.207	0.121	0.919	0.674
Temp. Scaling	0.037	0.018	0.147	0.902	0.197	0.045	0.626	0.672
MC Drop (p=0.5)	0.049	0.032	0.219	<u>0.904</u>	0.250	0.196	1.173	0.631
MC Drop (p=0.1)	0.059	0.044	0.290	0.881	0.191	0.131	0.754	0.726
CORAL	0.032	<u>0.026</u>	0.158	0.887	0.201	0.103	0.720	0.364
EnD ²	0.145	0.124	0.696	0.125	0.405	0.269	1.773	0.060
Scheme (a)	<u>0.030</u>	0.083	<u>0.151</u>	0.757	0.211	0.050	0.727	0.527
Scheme (b)	0.038	0.060	0.178	0.880	0.216	0.021	0.654	0.450
Scheme (c)	<u>0.030</u>	0.077	0.237	0.895	<u>0.186</u>	0.038	0.583	0.635
Scheme (d)	0.035	0.096	0.229	0.890	0.194	0.090	0.690	0.625
Scheme (e)	0.039	0.063	0.169	0.897	0.242	0.059	0.809	0.588
DomED (Teachers)	0.139	0.032	0.224	0.879	0.265	0.152	1.042	0.549
DomED (Ours)	0.029	0.061	0.158	0.936	0.184	<u>0.028</u>	<u>0.609</u>	<u>0.694</u>

Table 18. Out-of-distribution detection performance across four datasets, evaluated using the ROC-AUC metric. For each dataset, OOD samples are drawn from the test domain, while in-distribution (ID) samples are drawn from the train domains.

<i>PACS</i>										
Model	Art Painting		Cartoon		Photo		Sketch		Avg.	
	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.73	–	0.69	–	0.57	–	0.89	–	0.71	–
Ensemble	0.75	0.71	0.73	0.76	0.54	0.46	0.90	0.91	0.73	0.71
DomED	0.69	0.72	0.72	0.77	0.57	0.47	0.84	0.93	0.71	0.72
<i>OfficeHome</i>										
Model	Art		Clipart		Product		Real World		Avg.	
	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.72	–	0.79	–	0.62	–	0.57	–	0.67	–
Ensemble	0.75	0.65	0.81	0.84	0.63	0.68	0.58	0.65	0.69	0.70
DomED	0.69	0.71	0.72	0.79	0.51	0.66	0.50	0.64	0.60	0.70
<i>VLCS</i>										
Model	Caltech101		LabelMe		SUN09		VOC2007		Avg.	
	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.32	–	0.43	–	0.66	–	0.65	–	0.51	–
Ensemble	0.33	0.23	0.45	0.51	0.67	0.72	0.66	0.71	0.53	0.54
DomED	0.48	0.51	0.44	0.58	0.69	0.83	0.67	0.85	0.57	0.69
<i>TerraIncognita</i>										
Model	Location 100		Location 38		Location 43		Location 46		Avg.	
	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc	T.Unc	K.Unc
ERM	0.74	–	0.70	–	0.76	–	0.79	–	0.75	–
Ensemble	0.79	0.80	0.76	0.87	0.74	0.60	0.85	0.90	0.78	0.79
DomED	0.75	0.66	0.74	0.86	0.74	0.60	0.83	0.87	0.77	0.75